



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

I/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

Under Choice Based Credit System

GROUP-A (CIVIL, EEE & MECHANICAL)

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1101	English – I *	3	3	1	--	4	30	70	100
B17 BS 1102	Mathematics – I *	3	3	1	--	4	30	70	100
B17 BS 1105	Engineering Chemistry	3	3	1	--	4	30	70	100
B17 ME 1101	Engineering Mechanics	3	3	1	--	4	30	70	100
B17 ME 1102	Engineering Drawing	3	1	--	3	4	30	70	100
B17 CE 1101	Environmental Studies *	2	2	1	--	3	30	70	100
B17 BS 1107	Engineering Chemistry Lab	2	--	--	3	3	50	50	100
B17 BS 1108	English Communication Skills Lab – I *	2	--	--	3	3	50	50	100
B17 BS 1109	Engineering Workshop & IT Workshop	2	--	--	3	3	50	50	100
B17 BS 1111	Inner Engineering	--	--	--	2	2	--	--	--
Total		23	15	5	14	34	330	570	900

* Common to both Group - A and Group - B

Code: B17 BS 1101

ENGLISH - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop the communication skills of the students in both the formal and informal situations.
4. To expose the students to various forms of writing in formal settings in order to build confidence in standard grammar
5. To introduce the students various types of reading comprehension practices.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand the rudiments of LSRW Skills, comprehension and fluency of speech.
2. Gain confidence and competency in vocabulary and grammar.
3. Listen, speak, read and write effectively in both the academic and non- academic environment.
4. Extend his/her reading skills towards literature.
5. Strengthen his/her analytical and compositional skills.

SYLLABUS

Life through Language: An Effective Learning Experience

Life through Language has a systematic structure that builds up communicative ability progressively through the chapters. It will enable the learner to manage confusion; frame question for themselves and others; develop new ideas; support ideas with evidence; express themselves with poise and clarity; and think critically. Acquisition of skill leads to confidence.

UNIT-I

People and Places:- Word search - Ask yourself-Self-assessment-I -Self-assessment-II - Sentence and its types- Describing people, places and events-Writing sentences-Self-awareness-Self-motivation, Dialogue writing.

UNIT-II

Personality and Lifestyle:- Word quiz – Verbs-Adverbs-Negotiations-Proving yourself-Meeting Carl Jung- Describing yourself- Living in the 21st century- Using your dictionary-Communication-Adaptability.

UNIT-III

Media and Environment: - A list of 100 basic words – Nouns- Pronouns- Adjectives-News report- Magazine article- User's Manual for new iPod- A documentary on the big cat- Why we need to save our tigers: A dialogue- Global warming- Paragraph Writing-Arguing a case-Motivation- Problem solving.

UNIT-IV

Entertainment and Employment:- One word substitutes- Parts of speech- Gerunds and infinitives- An excerpt from a short story an excerpt from a biography- A consultant interviewing employees- Your first interview- Reality TV- Writing an essay-Correcting sentences- Integrity Sense of humor.

UNIT-V

Work and Business:- A list of 100 difficult words- Articles, Quantifiers- Punctuation - Open letter to the Prime Minister Business dilemmas: An email exchange- A review of *IPL: The Inside Story*, Mark Zuckerberg: World's Youngest Billionaire- A conversation about a business idea- Pair work: Setting up a new business- Recession- Formal letters-Emails- Reports- Professionalism-Ethics, Fill in the blanks.

Text Book:

1. *Life through Language: A Holistic Approach to Language Learning*. Board of Editors, Pearson Publishers, India. 2013.

Reference Books:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008.
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008.
3. Know Your English (Volume 1&2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Mathukutty Monippally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that is essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Learn about linear first order and higher order ordinary differential equations and their applications.
4. Acquire knowledge of Laplace transform, partial differentiation and their applications.
5. Learn certain first order and higher order partial differential equations.

Course Outcomes: At the end of the Course, Student will be able to:

1. Solve linear ordinary differential equations of first order and first degree. Also will be able to apply the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.
2. Solve linear ordinary differential equations of second order and higher order. Also will be able to apply the knowledge in simple applications such as LCR circuits and Simple harmonic motion.
3. Determine Laplace transform and inverse Laplace transform of various functions.
4. Use Laplace transforms to solve a linear ODE.
5. Calculate total derivative, Jacobian and maxima/minima of functions of two variables.
6. Form partial differential equations and solve some standard types of first order PDEs. Find complimentary function and particular integral of linear higher order homogeneous and non-homogeneous PDEs.

SYLLABUS

UNIT I: Differential equations of first order and first degree:

Linear, Bernoulli, Exact, Reducible to exact types.

Applications: Newton's Law of cooling, Law of natural growth and decay, Orthogonal trajectories, Simple electrical circuits, Chemical reactions.

UNIT II: Linear differential equations of higher order:

Non-homogeneous equations of higher order with constant coefficients with RHS term of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax}V(x)$, $xV(x)$, Method of Variation of parameters.

Applications: LCR circuit, Simple Harmonic motion.

UNIT III: Laplace transforms:

Laplace transforms of standard functions, transforms of $tf(t)$, $f(t)/t$, properties, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function, Inverse Laplace transforms, convolution theorem (without proof).

Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms.

UNIT IV: Partial differentiation:

Introduction, Homogeneous functions, Euler's theorem, Total derivative, Chain rule, which variable is to be treated as constant, Functional dependence, Jacobians, Taylor series for a function of two variables, Leibnitz rules for differentiation under the integral sign.

Applications: Errors and Approximations, Maxima and Minima of functions of two variables without constraints, Lagrange's method (with constraints)

UNIT V: First order and higher order partial differential equations:

Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of first order Lagrange linear equation and nonlinear equations of standard types (excluding Charpit's method).

Solutions of Linear homogeneous and non-homogeneous Partial differential equations with constant coefficients - RHS terms of the type e^{ax+by} , $\sin(ax+by)$, $\cos(ax+by)$, $x^m y^n$.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
2. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson.
3. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
4. Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING CHEMISTRY
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Knowledge of basic concepts of chemistry for engineering students will help them as professional engineers later in design and material selection as well as utilizing the available resources.

Course Objectives:

1. To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose.
2. To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc.
3. To understand the concepts of galvanic cells and corrosion with theories like electro chemical theory.
4. To understand the importance of water.
5. To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries.

Course Outcomes:

1. At the end of the course the students learn the advantages and limitations of plastic materials and their use in design.
2. Fuels which are used commonly and their economics, advantages and limitations are discussed.
3. Students gained Knowledge reasons for corrosion and some methods of corrosion control.
4. Students understands the impurities present in raw water, problems associated with them and how to avoid them.
5. Similarly students understand liquid crystals and semi conductors. Students can gain the building materials , solar materials, lubricants and energy storage devices.

SYLLABUS

UNIT-I: High Polymers and Plastics; Rubbers & Elastomers:

Polymerization Definition, Types of Polymerization, Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6, Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers, Biodegradable Polymers - PHBV, Nylon 2, Nylon 6. Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N;

UNIT-II :Fuel Technology& Lubricants:

Fuels: - Introduction – Classification of fuels, Calorific value – HCV and LCV, Determination of Calorific value by bomb calorimeter; Proximate and ultimate analysis of coal, coke: manufacture of coke by Otto – Hoffmann's by-product coke oven process; Refining of Petroleum, Knocking-octane number of gasoline, cetane number of diesel oil. Synthetic Petrol; LPG , CNG.

Lubricants:- Definition, Mechanism of Lubrication, Properties of Lubricants (Definition and significance)

UNIT-III: Electrochemical cells and Corrosion:

Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion- material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electroless plating ; Paints.

UNIT-IV: Water technology:

Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods.

UNIT-V: Chemistry of Engineering Materials& Advanced Engineering materials

Cement:- Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete.

Refractories: - Definition, Characteristics, classification, Properties and failure of refractories.

Solar Energy: - Construction and working of Photovoltaic cell, applications.

Solid State Materials: Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors.

Liquid Crystals: - Definition – Classification with examples – Applications

Text Books:

1. Engineering Chemistry by Jain and Jain, Dhanpat Rai publishing co.
2. Engineering Chemistry by Willy India Pvt Ltd.
3. Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy; Silicon Publications.

Reference Books:

1. Engineering Chemistry by Shikha Agarwal; Cambridge University Press,2015 edition.
2. A text of Engineering Chemistry by S.S.Dara; S.Chand & Co Ltd.

ENGINEERING MECHANICS
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To impart the basic concepts of Engineering Mechanics and the principles of various force systems under static and dynamic conditions.
2. Comprehensive, theory based understanding of the underpinning natural and physical sciences and the engineering fundamentals applicable to the engineering discipline.
3. Application of established engineering methods to complex engineering problem solving.
4. The course is the key prerequisite course to sequences of courses dealing with mechanics of machines, stress analysis and design of mechanical systems.

Course Outcomes:

Students will be able to

1. Determine the resultant of the given force systems.
2. Analyze force systems using equations of equilibrium.
3. Determine centroid, center of gravity and moment of inertia of areas and bodies.
4. Analyze trusses and simple beams.
5. Distinguish between kinematics and kinetics.
6. Apply the work energy and impulse momentum methods of various engineering problems.

SYLLABUS

UNIT-I

Basic Concepts:

Scalar and vector quantities- Representation of vectors- Free vector force, Specification of force- Effect of force on rigid body- Free body diagram. Concurrent Forces in a plane: Principles of statics-Resolution and Composition of forces in a plane-Equilibrium of concurrent forces in a plane- Method of projections- Equilibrium of three forces in a plane Method of moments. Parallel Force system in a plane.

UNIT-II

Centroid & Moment of Inertia: Centroid & M.I – Area & Mass M.I – Radius of Gyration, Parallel axis– Perpendicular axis theorem – Simple Problems.

UNIT-III

General Case of Forces in a Plane: Resultant and equilibrium of general case of forces in a plane, Statically determinate plane trusses-Method of joints and Method of sections.

Friction – Coulombs laws of dry friction – Limiting friction, Problems on Wedge friction, Belt Friction-problems.

UNIT-IV

Dynamics of Particles - Rectilinear Motion – Kinematics, D'Alembert's principle, Kinetics – Work & Energy – Impulse Moment, Direct Central Impact – coefficient of restitution.

Curvilinear Motion – Kinematics, Projectile Motion, Moment of momentum, Work & Energy in Curvilinear motion.

UNIT-V

Dynamics of Rigid Bodies - Rigid body rotation – Kinematics - Kinetics – Work & Energy in Rigid body rotation, Plane Motion – Kinematics – Instantaneous center of rotation, Kinetics - Work-Energy principle in plane motion.

Text Book:

1. Engineering Mechanics by S.Timoshenko and D.HYoung McGraw-Hill.

Reference Books:

1. Engineering Mechanics, Vol.1 & 2 by J.L. Meriems and L.G. Kraige.
2. Engineering Mechanics by Singer.
3. Engineering Mechanics by K.L. Kumar, Tata Mc-Graw Hill.
4. Engineering mechanics by Bhavikatti, New age international.

ENGINEERING DRAWING
(Common to CIV, EEE & ME)

Lecture	: 1 Period	Int.Marks	: 30
Practice	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

Course Outcomes:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent the object in 3D view through isometric views.
5. Convert the isometric view to orthographic view and vice versa.

SYLLABUS

UNIT I

Polygons: Constructing regular polygons by general methods, inscribing and describing polygons on circles.

Curves: Parabola, Ellipse and Hyperbola by general methods, cycloids, involutes, tangents & normals for the curves.

UNIT II

Orthographic Projections: Horizontal plane, vertical plane, profile plane, importance of reference lines, projections of points in various quadrants, projections of lines, lines parallel either to one of the reference planes (HP, VP or PP)

Projections of straight lines inclined to both the planes, determination of true lengths, angle of inclination and traces- HT, VT

UNIT III

Projections of planes: regular planes perpendicular/parallel to one plane and inclined to the other reference plane; inclined to both the reference planes.

UNIT IV

Projections of Solids – Prisms, Pyramids, Cones and Cylinders with the axis inclined to one of the planes.

UNIT V

Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Text Books:

1. Engineering Drawing by N.D. Butt, Chariot Publications.
2. Engineering Drawing by Agarwal & Agarwal, Tata McGraw Hill Publishers

Reference Books:

1. Engineering Drawing by K.L.Narayana & P. Kannaiah, Scitech Publishers.
2. Engineering Graphics for Degree by K.C. John, PHI Publishers.
3. Engineering Graphics by P. Varghese, McGrawHill Publishers.
4. Engineering Drawing + AutoCad – K Venugopal, V. Prabhu Raja, New Age

ENVIRONMENTAL STUDIES
(Common to all Branches)

Lecture : 2 Periods	Int.Marks : 30
Tutorial : 1 Period.	Ext. Marks : 70
Exam : 3 Hrs.	Credits : 2

Course Objectives:

1. Developing an awareness and sensitivity to the total environment and its related problems.
2. Motivating students for active participation in environmental protection and improvement.
3. Developing skills for active identification and development of solutions to environmental problems
4. Evaluation of environment programmes in terms of social, economic, ecological and aesthetic factors.
5. Creating a “CONCERN AND RESPECT FOR THE ENVIRONMENT”.

Course Outcomes:

1. To bring awareness among the students about the nature and natural ecosystems
2. Sustainable utilization of natural resources like water, land, energy and air
3. Resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation
4. Safe guard against industrial accidents particularly nuclear accidents
5. Constitutional provisions for the protection of natural resources

SYLLABUS

UNIT – I

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness. Global Environmental Crisis

Ecosystems:

Basic Concepts - Structure and Functions of Ecosystems: Producers, Consumers and Decomposers. Types of Ecosystems: Forest Ecosystems, Grassland Ecosystems Desert Ecosystems and Aquatic Ecosystems

UNIT-II

Biodiversity:

Introduction to Biodiversity, Values of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Hotspots of Biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of Biodiversity.

UNIT-III

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture. **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people. **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems. **Energy Resources:** Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

UNIT-IV

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and Global warming, Acid rain and Ozone layer depletion. Solid Waste Management: Composting, Vermiculture, Urban and Industrial Wastes, Recycling and Reuse.

Environmental Problems in India:

Drinking water, Sanitation and Public health, Population growth and Environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

UNIT-V

Institutions and Governance:

Regulations by Government- Environmental Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act. Environmental Impact Assessment (EIA)

Case Studies:

Chipko Movement, Narmada Bachao Andolan, Silent Valley Project, Mathura Refinery and Taj Mahal, Industrialization of Patancheru, Nuclear reactor at Nagarjuna Sagar, Tehri Dam, Ralegaon Siddhi (Anna Hazare), Kolleru lake – Aquaculture, Fluorosis in Andhra Pradesh & Telangana.

Field Work:

Visit to a local area to document and mapping environmental assets. Visits to Industries, Water Treatment Plants, Effluent Treatment Plants.

Text Books:

1. Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada
2. Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford University Press.
3. Environmental Studies, Dr. M. Sita Rama Reddy, Dr. K. Anji Reddy, Silicon Publications, ASR Nagar, Bhimavaram
4. Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula Rani; Pearson Education, Chennai

Reference Books:

1. Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2. A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3. Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4. Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

**ENGINEERING CHEMISTRY LAB
(Common to CIV, EEE & ME)****Lab : 3 Periods**
Exam : 3 Hrs.**Int.Marks : 50**
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To investigate and understand Physical behavior in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world.
2. Effectively communicate experimental results and solutions to application problems through oral and written reports.
3. Understand the basic concepts, definitions, characteristics and phenomena.
4. Recognize the classical ideas and chemical phenomena and also define and analyse the concepts.

Course Outcomes:

1. An understanding of Professional and develop confidence on recent trends.
2. Able to gain technical knowledge of measuring, operating and testing of chemical instruments and equipments.
3. Acquire ability to apply knowledge of chemistry.
4. Exposed to the real time working environment.
5. Demonstrate the ability to learn Principles, design and conduct experiments.
6. Ability to work on laboratory and multidisciplinary tasks.

LIST OF EXPERIMENTS**Introduction to chemistry Laboratory**

1. Estimation of HCl using standard Sodium Hydroxide.
2. Determination of total hardness of water by EDTA method.
3. Estimation of Ferrous Iron by KMnO_4 .
4. Estimation of oxalic acid by KMnO_4
5. Estimation of Mohr's salt by $\text{K}_2\text{Cr}_2\text{O}_7$
6. Estimation of Dissolved oxygen by Winkler's method.
7. Determination of pH by pH meter and universal indicator method.
8. Conductometric titration of strong acid Vs strong base
9. Conductometric titration of strong acid Vs weak base.
10. Potentiometric titration of strong acid Vs strong base
11. Potentiometric titration of strong acid Vs weak base
12. Preparation of Phenol formaldehyde resin.
13. Determination of saponification value of oils
14. Determination of pour and cloud points of lubricating oil.
15. Determination Acid value of oil.

Text Books:

1. Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.Engineering College.

Reference Books:

1. Laboratory manual on Engineering Chemistry by Dr.Sudha Rani ;Dhanpat Rai Publishing Company.
2. Engineering Chemistry Laboratory manual – I & II by Dr. K.Anji Reddy; Tulip Publications.

ENGLISH COMMUNICATION SKILLS LAB- I
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn through practice the communication skills of listening, speaking, reading and writing.
2. To make students recognize the sounds of English through Audio- Visual aids.
3. To familiarize the students with stress and intonation.
4. To help the students build their confidence in speaking skills.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students improve their speaking skills in real contexts.
3. Students learn standard pronunciation and practice it daily discourse.
4. Students give up their communicative barriers.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. **'INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

**ENGINEERING WORKSHOP & IT WORKSHOP
(Common to CIV, EEE & ME)**

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

PART-A ENGINEERING WORKSHOP

Course Objective:

To impart hands-on practice on basic engineering trades and skills.

Course Outcomes:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Prepare jobs of various shapes using black smithy.
3. Make basic house wire connections.
4. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry	Fitting
1. T-Lap Joint 2. Cross Lap Joint 3. Dovetail Joint 4. Mortise and Tenon Joint	1. Vee Fit 2. Square Fit 3. Half Round Fit 4. Dovetail Fit
Black Smithy	Tin Smithy
1. Round rod to Square 2. S-Hook 3. Round Rod to Flat Ring 4. Round Rod to Square headed bolt	1. Taper Tray 2. Square Box without lid 3. Open Scoop 4. Funnel
House Wiring	
1. Parallel / Series Connection of three bulbs 2. Stair Case wiring 3. Florescent Lamp Fitting 4. Measurement of Earth Resistance	

Note: At least two exercises to be done from each trade.

Reference:

1. Elements of workshop technology, Vol.1 by S. K. and H. K. Choudary.

PART B: IT WORKSHOP:

Course Objectives:

1. Understand the basic components and peripherals of a computer.
2. To become familiar in configuring a system.
3. Learn the usage of productivity tools.
4. Acquire knowledge about the netiquette and cyber hygiene.
5. Get hands on experience in trouble shooting a system

Course Outcomes:

At the end of the course the students can able to

1. Assemble and disassemble the systems
2. Use the Microsoft office tools
3. Install various software
4. Know about various search engines
5. Trouble shoot various Hardware and Software problems
6. Use MATLAB and LATEX softwares

LIST OF EXERCISES

1. System Assembling, Disassembling and identification of Parts / Peripherals
2. Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.
3. MS-Office / Open Office
 - a) Word - Formatting, Page Borders, Reviewing, Equations, symbols.
 - b) Spread Sheet - organize data, usage of formula, graphs, charts.
 - c) Power point - features of power point, guidelines for preparing an effective presentation.
 - d) Access- creation of database, validate data.
4. Network Configuration & Software Installation-Configuring TCP/IP, proxy and firewall settings. Installing application software, system software & tools.
5. Internet and World Wide Web-Search Engines, Types of search engines, netiquette, cyber hygiene.
6. Trouble Shooting-Hardware trouble shooting, Software trouble shooting.
7. MATLAB- basic commands, subroutines, graph plotting.
8. LATEX-basic formatting, handling equations and images.

Reference Books:

1. Computer Hardware, Installation, Interfacing, Troubleshooting and Maintenance, K.L. James, Eastern Economy Edition.
2. Microsoft Office 2007: Introductory Concepts and Techniques, Windows XP Edition By Gary B. Shelly, Misty E. Vermaat and Thomas J. Cashman (2007, Paperback).
3. LATEX- User's Guide and Reference manual, Leslie Lamport, Pearson, LPE, 2/e.
4. Getting Started with MATLAB: A Quick Introduction for Scientists and Engineers, Rudraprathap, Oxford University Press, 2002.
5. Scott Mueller's Upgrading and Repairing PCs, 18/e, Scott. Mueller, QUE, Pearson, 2008
6. The Complete Computer upgrade and repair book, 3/e, Cheryl A Schmidt, Dreamtech.
7. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dreamtech.
8. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.

**INNER ENGINEERING
(Common to CIV, EEE & ME)**

Practice : 2 Periods

Course Objectives:

Student should learn:

1. Human values and tools to lead a happy, stress-free life.
2. Yoga asanas, Pranayama, Sudarshan Kriya & Meditation
3. At-least two creative arts out of photography, sketching, craft-making, singing, clay molding, upcycling etc.
4. Concentration pranayama, Ego bursting process.
5. To take up responsibility for society and teach classes of their choice to school children.
6. About good food habits for good health.

Course Outcomes:

Student should be capable of

1. To improve his concentration levels and improve his public speaking abilities.
2. To balance his academic and non-academic activities (Work Life Balance).
3. To widen his vision and increase his breadth of perspective in his journey of 4 years.
4. To improve his communications skills, leadership, teamwork and decision-making abilities.
5. To inculcate creativity & innovation, planning & organizing as part of their life.
6. Taking responsibility for themselves and people around them.
7. To make their journey more fun and enjoyable.

SYLLABUS

UNIT-I

YES!+ Workshop:

Yoga Postures – Seven Layers To our Existence – Puzzles – Sources Of Energy – Live in the present Moment – Importance of Breath – Ujjai Breath – Pranayama – Sudarshana Kriya.

UNIT-II

YES!+ Workshop:

Yoga Postures (Suryanamaskars) – Giving 100% in everything – Time management – Happiness point – Opposite Values – Pranayama – Sudarshan kriya

UNIT-III

YES!+ Workshop:

Yoga Postures – Knowledge points (Acceptance, opinions discretion and handling mistakes) – Eye Gazing Process – Dance – Life Story process – Sudarshana Kriya (short) – Eternal life – Ego Bursting – Relationships – Parents – Studies – Compliments/Praising process.

UNIT-IV

Creative Arts:

Photography – Sketching – Handy-crafts – Clay molding – Singing – Upcycling – Communing with nature – Creative writing.

UNIT -V

Service:

Leadership in action – Contributing to society – Take up Responsibility –Empowerment – Public Speaking – Art of Teaching.

Reference Books:

1. Discourse on Patanjali Yoga Sutras by H.H Sri SriRavishankar
2. Human values and professional ethics byRRGaur,RSangal,GPBagaria
3. The Art of Photography by AL Judge
4. Hand made in India : Crafts of India by Ranjan Aditi

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R17)
I/IV B.TECH
 (With effect from **2017-2018** Admitted Batch onwards)
 Under Choice Based Credit System

GROUP-A (CIVIL, EEE & MECHANICAL)
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1201	English – II *	3	3	1	--	4	30	70	100
B17 BS 1202	Mathematics – II	3	3	1	--	4	30	70	100
B17 BS 1203	Mathematics – III*	3	3	1	--	4	30	70	100
B17 BS 1204	Engineering Physics	3	3	1	--	4	30	70	100
B17 CS 1201	Computer Programming Using C	3	3	1	--	4	30	70	100
# DS 1	Department Subject	3	3	1	--	4	30	70	100
B17 BS 1206	Engineering Physics Lab	2	--	--	3	3	50	50	100
B17 BS 1208	English Communication Skills Lab – II *	2	--	--	3	3	50	50	100
B17 CS 1204	C Programming Lab	2	--	--	3	3	50	50	100
B17 BS 1210	Engineering Physics Virtual Labs- Assignments	--	--	--	2	2	--	--	--
B17 BS 1211	NCC	--	--	--	2	2	--	--	--
Total		24	18	6	13	37	330	570	900

*** Common to both Group - A and Group - B**

#DS 1	CIVIL	B17 CE 1201	Building Materials and Construction
	EEE	B17 EE 1201	Circuit Theory
	MECHANICAL	B17 EE 1202	Basic Electrical and Electronics Engineering

ENGLISH - II
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop spoken and written forms of the students in both formal and informal situations.
4. To enrich the vocabulary of the students through the usage of the lexis in appropriate contexts.
5. To expose them to the correct structural patterns of English.

Course Outcomes: At the end of the Course, Student will be able to:

1. To comprehend the speech of people belonging to different backgrounds and regions.
2. Understand the importance of speaking and writing for personal and professional communication and practice it in real contexts.
3. To express fluently and accurately in social discourse.
4. Participate in group activities like role-plays, discussions and debates.
5. Identify the discourse features, and improve intensive and extensive reading skills.

SYLLABUS

UNIT I:

- A. Detailed-Text: Unit 1: ' The Greatest Resource- Education'
- B. Non-Detailed Text: Lesson 1: ' A P J Abdul Kalam' from The Great Indian Scientists.

UNIT II:

- A. Detailed-Text: Unit 2: ' A Dilemma'
- B. Non-Detailed Text: Lesson 2:'C V Raman' from The Great Indian Scientists.

UNIT III:

- A. Detailed-Text: Unit 3: 'Cultural Shock': Adjustments to new Cultural Environments
- B. Non-Detailed Text: Lesson 3:'Homi Jehangir Bhabha' from The Great Indian Scientists.

UNIT IV:

- A. Detailed-Text: Unit 4: 'The Lottery'
- B. Non-Detailed Text: Lesson 4: 'Jagadish Chandra Bose' from The Great Indian Scientists.

UNIT V:

- A. Detailed-Text: Unit 5: ' The Chief Software Architect'
- B. Non-Detailed Text: Lesson 5: ' Prafulla Chandra Ray' from The Great Indian Scientists

Detailed Textbook:

1. English Encounters Published By Maruthi Publishers.

Non-Detailed Text Book:

1. The Great Indian Scientists Published by Cengage learning.

MATHEMATICS – II
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand some basic numerical methods to solve algebraic and transcendental equations.
4. Learn interpolation methods for equally spaced and unequally spaced data points. Learn methods for numerical evaluation of integrals and for solving first order ODEs.
5. Learn concepts of a Fourier series and Fourier transform.

Course Outcomes: At the end of the Course, Student will be able to:

1. Find a real root of algebraic and transcendental equations using different methods.
2. Know the relation between the finite difference operators. Determine interpolation polynomial for a given data.
3. Evaluate numerically certain definite integrals applying Trapezoidal and Simpson's rules.
4. Solve a first order ordinary differential equation by Euler and RK methods.
5. Find Fourier series of a given function satisfying Dirichlet conditions. Find half range cosine and sine series for appropriate functions.
6. Find Fourier transforms Fourier cosine and sine transforms of appropriate functions and evaluate certain integrals using inverse transforms and Fourier integral.

SYLLABUS

UNIT I: Solution of Algebraic and Transcendental Equations:

Introduction, Bisection method, Method of false position, Iteration method, Newton- Raphson method (One variable and simultaneous Equations).

UNIT II: Interpolation:

Introduction, Errors in polynomial interpolation, Finite differences, Forward differences, Backward differences, Central differences and Symbolic relations between the operators, Differences of a polynomial, Newton's formulae for interpolation, Interpolation with unequal intervals, Lagrange's interpolation formula.

UNIT III: Numerical Integration and solution of Ordinary Differential equations:

Trapezoidal rule, Simpson's $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rules, Solution of ordinary differential equations by Taylor series method, Picard's method of successive approximations, Euler's method, Runge-Kutta methods (second order and fourth order).

UNIT IV: Fourier Series:

Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Even and odd functions, Change of interval, Half-range sine and cosine series, Parseval's formula.

UNIT V: Fourier Transforms:

Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, properties, inverse transforms, Parseval's identities, Finite Fourier transforms.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
2. V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
4. David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press.
5. Srimanta Pal, Subodh C. Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

MATHEMATICS - III
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand important concepts of Linear algebra including solution of linear simultaneous equations, eigen values and eigen vectors.
4. Learn Beta, Gamma functions and how to evaluate double and triple integrals.
5. Learn fundamental concepts of vector calculus that help understand other Engineering courses.

Course Outcomes: At the end of the Course, student will be able to:

1. Determine rank, and solve a system of linear simultaneous equations numerically using various matrix methods.
2. Determine Eigen values and Eigen vectors of a given matrix, Reduce a Quadratic form to its canonical form and classify.
3. Evaluate double integrals over a region and triple integral over a volume.
4. Use the knowledge of Beta and Gamma functions in evaluation of different integrals.
5. Find gradient of a scalar function, divergence and curl of a vector function. Use vector identities for solving problems.
6. Evaluate line, surface and volume integrals by the use of Green's, Stokes' and Gauss divergence theorems.

SYLLABUS

UNIT I: Linear systems of equations:

Rank, Echelon form, Normal form, Solution of linear systems, Gauss elimination, Gauss-Jordan, Jacobi and Gauss-Seidel methods.

Applications: Finding the current in electrical circuits.

UNIT II: Eigen values - Eigen vectors and Quadratic forms:

Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix by using Cayley-Hamilton theorem, Diagonalization, Quadratic forms, Reduction of a Quadratic form to Canonical form, Rank, Positive, Negative, Semi-Definite and indefinite forms of a Quadratic form, Index and Signature of a Quadratic form.

Applications: Free vibration of a two-mass system.

UNIT III: Multiple integrals:

Double and triple integrals, Change of variables, Change of order of integration. Application to finding Areas, Moment of Inertia and Volumes.

Beta and Gamma functions, Properties, Relation between Beta and Gamma functions, Application to evaluation of improper integrals. The error function and the complimentary error function.

UNIT IV: Vector Differentiation:

Gradient, directional derivative, Divergence, Curl, Incompressible flow, solenoidal and irrotational vector fields, second order operators, vector identities.

UNIT V: Vector Integration:

Line integral, Work done, Potential function; Area, Surface and volume integrals, Flux, Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof) and related problems.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson edn.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
3. Peter O'Neil, Advanced Engineering Mathematics, 7th edition, Cengage Learning.
4. D.W. Jordan and T. Smith, Mathematical Techniques, Oxford University Press.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING PHYSICS
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

To re-orient Physics curriculum to the needs of the circuit and non circuit branches of Engineering / Technology courses offered by SRKREC(A) Bhimavaram that serves to understand the general and branch specific basic topics and introduce the related advanced technologies to the stake holders.

1. To impart the knowledge of physical optics phenomena like Interference and Diffraction required to understand the design and working of optical instruments with high resolution.
2. To understand the concept of coherence and generation of high intensity light sources and propagation of light waves in fibers for technological applications.
3. To study and analyze the behavior of electric and magnetic fields and their mutual interaction, and to understand the science of ultrasonics.
4. To understand the basic concepts of Quantum mechanics to know the behavior of fundamental particles in materials and to understand the basic nature of materials in general and classify them.
5. To study the structure property relationship of crystalline solids and understand basics of nanotechnology.

Course Outcomes: At the end of the course the students-

1. Learn the basic concepts of interference and diffraction of light and its applications.
2. Understand the science of producing high intensity light beams for technological applications and also understand the propagation of light waves in optical fibers in various applications.
3. Understand the inter relationship of electric and magnetic fields and learn ultrasonic's as a tool for technological applications
4. Learn the behavior of particles at the very microscopic level by using wave nature of particles and understand the behavior of materials and be able to classify them using the band theory of solids.
5. Learn the basics of structures of solid materials and nano material preparation Techniques/methods.

SYLLABUS

UNIT I: Interference and Diffraction

Principle of superposition-coherence-interference in thin films (reflected system) – Wedge shaped film-Newton's rings-Michelson's interferometer. Fraunhofer's diffraction at single slit, Diffraction grating-Resolving power of a grating.

UNIT- II: Lasers and Optical Fibers

Introduction, Spontaneous emission and Stimulated emission – Einstein's relation – Requirements of Laser device- Ruby laser- He-Ne gas laser- Characteristics of laser- Applications.

Description of optical fiber, Principle of light propagation- Optical fiber –Acceptance angle-Numerical aperture of optical fiber- Modes of propagation- Classification of fibers- Applications of fiber.

UNIT- III: Electro Magnetic Fields and Ultrasonics

Concept of Electromagnetic induction , Faraday's law, Lenz's law, Electric fields due to time varying magnetic fields, Magnetic fields due to time varying electric fields, Displacement current, Modified Ampere's law, Maxwell's equations and their significance (without derivation).

Definition of Ultrasonics-Methods of Producing Ultrasonics- Detection of Ultrasonics- Applications of Ultrasonics.

UNIT- IV: Quantum Mechanics and Band Theory of Solids

Introduction, de Broglie matter waves- properties-Experimental confirmation, wave function-significance- Schrodinger's time dependent and time independent wave equations- Eigen values and functions, Particle in a box.

Band theory of Solids- Introduction- Kronig Penney model (Qualitative)- Energy bands of crystalline solids- Distinction between Conductors, Semi conductors and insulators.

UNIT-V: Crystallography and Nano Materials

Basis and Lattice, Crystal systems, Bravais lattice, Unit cell Coordination number – Packing fraction for SC ,FCC, and BCC lattices, Miller indices- Diffraction of X rays from crystals- Bragg's law.

Introduction to Nanomaterials – Synthesis methods : Condensation, ball milling, sol-gel, chemical vapour deposition methods, properties and applications.

Text Books:

1. Physics by Resnick & Halliday. Wiley - Eastern (India) Ltd.
2. Engineering Physics By M.N. Avadhanulu & PG Kshirasagar, S.Chand & Co.
3. Engineering Physics by V. Rajendran, Mc Graw Hill Education (India) Pvt.Ltd.

Reference Books:

1. Engineering Physics by MR Srinivasan, New International Publishers.
2. Solid State Physics – A.J. Dekkar, MacMillan (India) Ltd.
3. Engineering Physics by Gaur Gupta, Dhanpat Rai Publications, Meerut, India.
4. Engineering Physics by P k Palanisamy, Scitech Publications (India) Pvt.Ltd

(Note: Assignment Marks of **Engineering Physics** are to be considered from the Internal marks of **Engineering Physics-- Virtual Labs – Assignments B17 BS 1210**)

COMPUTER PROGRAMMING USING C
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Formulating algorithmic solutions to problems and implementing algorithms in C

1. Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux
2. understanding branching, iteration and data representation using arrays
3. Modular programming and recursive solution formulation
4. Understanding pointers and dynamic memory allocation
5. Understanding miscellaneous aspects of C
6. Comprehension of file operations

Course Outcomes:

1. Understand the basic terminology used in computer programming
2. Write, compile and debug programs in C language.
3. Use different data types in a computer program.
4. Design programs involving decision structures, loops and functions.
5. Explain the difference between call by value and call by reference
6. Understand the dynamics of memory by the use of pointers
7. Use different data structures and create/update basic data files.

SYLLABUS

UNIT I:

Unit objective: Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux

Introduction: Computer systems, Hardware and Software Concepts.

Problem Solving: Algorithm / Pseudo code, flowchart, program development steps, computer languages: machine, symbolic and highlevel languages, Creating and Running Programs: Writing, Editing(vi/emacs editor), Compiling(gcc), Linking and Executing in under Linux.

BASICS OF C: Structure of a c program, identifiers, basic data types and sizes. Constants, Variables, Arithmetic , relational and logical operators, increment and decrement operators, conditional operator, assignment operator, expressions, type conversions, Conditional Expressions, precedence and order of evaluation, Sample Programs.

UNIT II:

Unit objective: understanding branching, iteration and data representation using arrays

SELECTION – MAKING DECISION: TWO WAY SELECTION: if-else, null else, nested if, examples, Multi-way selection: switch, else-if, examples.

ITERATIVE: loops- while, do-while and for statements , break, continue, initialization and updating, event and counter controlled loops, Looping applications: Summation, powers, smallest and largest.

ARRAYS: Arrays- concepts, declaration, definition, accessing elements, storing elements, Strings and String Manipulations, 1-D arrays, 2-D arrays and character arrays, string manipulations, Multidimensional arrays, array applications: Matrix operations, checking the symmetricity of a Matrix. **STRINGS: concepts, c strings.**

UNIT III:

Objective: Modular programming and recursive solution formulation

FUNCTIONS- MODULAR PROGRAMMING: functions, basics, parameter passing, storage classes extern, auto, register, static, scope rules, block structure, user defined functions, standard library functions, recursive functions, Recursive solutions for fibonacci series, towers of Hanoi, header files, C Preprocessor, example c programs, Passing 1-D arrays, 2-D arrays to functions.

UNIT IV:

Objective: Understanding pointers and dynamic memory allocation

POINTERS: pointers- concepts, initialization of pointer variables, pointers and function arguments, passing by address- dangling memory, address arithmetic, character pointers and functions, pointers to pointers, pointers and multi-dimensional arrays, dynamic memory management functions, command line arguments

UNIT V:

Objective: Understanding miscellaneous aspects of C

ENUMERATED, STRUCTURE AND UNION TYPES: Derived types- structures declaration, definition and initialization of structures, accessing structures, nested structures, arrays of structures, structures and functions, pointers to structures, self referential structures, unions, typedef, bit-fields, program applications

BIT-WISE OPERATORS: logical, shift, rotation, masks.

Objective: Comprehension of file operations

FILEHANDLING: Input and output- concept of a file, text files and binary files, Formatted I/O, File I/O operations, example programs

Text Books:

1. Problem Solving and Program Design in C, Hanly, Koffman, 7thed, PERSON
2. Programming in C, Second Edition PradipDey and Manas Ghosh, OXFORD Higher Education
3. Programming in C, A practical approach Ajay Mittal PEARSON
4. The C programming Language by Dennis Richie and Brian Kernighan
5. Programming in C, B. L. Juneja, Anith Seth, Cengage Learning.

Reference Books:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming with C, Bichkar, Universities Press
3. Programming in C, ReemaThareja, OXFORD
4. C by Example, Noel Kalicharan, Cambridge

BUILDING MATERIALS AND CONSTRUCTION (For CIVIL)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To promote development, production, standardization and large-scale application of cost-effective innovative building materials and construction technologies in housing and building sector.
2. To develop and promote methodologies and technologies related to different Building constructions
3. To promote new building materials and components through technical support and encouraging entrepreneurs to set up production units.
4. To provide standard scientific and technical services to professionals, construction agencies and entrepreneurs in selection, evaluation and design engineering materials.

Course Outcomes:

1. Define and classify various stones, clay products used in construction sector.
2. Find various types of wood their conversion and relevant BIS testing procedures to be carried out to ascertain the quality of building materials.
3. Identify the major ingredients of construction materials like lime, cement, aggregate and their use in the construction industry.
4. Select different materials for finishing's and various constructions pertaining to masonry works and foundations.
5. Develop the conceptual knowledge of various supports in building construction.

SYLLABUS

UNIT I: STONES, BRICKS AND CLAY PRODUCTS

Stones: Classification of stones, Properties of building stones, Stone quarrying, precautions in blasting **Bricks:** Classification of Bricks, Manufacture of Bricks, general qualities of Bricks as per IS code, tests for good bricks as per IS code, including field tests. **Clay Products:** Tiles-types, manufacturing and their uses, Earth-ware, Terra-cotta, stone ware, Porcelain.

UNIT II: WOOD, WOOD BASED PRODUCTS

Wood: cross section details of trees, their general properties, characteristics of good timber defects in timber, mechanical properties of timber, seasoning and its importance, Decay of timber, **Wood based Products:** Veneers, Plywood and its types, Manufacturing of plywood, plywood grades as per IS code, Laminated wood, merits of plywood and laminated wood, Lamin Boards, Block boards, Batten board, Particle boards

UNIT III: LIME, CEMENT & AGGREGATES

Lime: Various ingredients of lime, Constituents of lime stone, classification of lime ,
Cement: Natural and artificial cements, types of artificial cements and their uses, Wet and dry process of manufacturing ordinary Portland cement (OPC), composition of cement, Various field and Laboratory tests on OPC as per IS code, Storage of cement. **Aggregates:** Classification of aggregate – Coarse and fine aggregates, Particle shape and Texture, Specific gravity, Bulk density, porosity and Absorption, Moisture content of Aggregate – Bulking of sand, Sieve analysis.

UNIT IV: FINISHINGS, MASONRY AND FOUNDATIONS

Finishings: Paints and Varnishes: Constituents and characteristics of paints, types of paint and their uses, painting defects, causes and remedies. Constituents of varnishes, types of varnish and their uses, Pointing and Plastering. **Masonry:** Different types of Stone Masonry- Plan, Elevation, Sections of stone Masonry works- Brick Masonry- Different Types of Bonds- Plan, Elevation and section of Brick Bonds upto Two-Brick wall thickness- Partition walls- Different types of Block Masonry- Hollow concrete Blocks- FAL-G Blocks, Hollow Clay Blocks. **Foundations:** Types- strip, isolated, strap, combined footings, Raft-Mat- flat slab and Beam raft, box type raft.

UNIT V: ROOFING, FORM WORK & SCAFFOLDING

Roofing: Mangalore tiled roof, RCC roof, Madras terrace roof, Hollow tiled roof, Asbestos cement, Fibre glass, Aluminum G.I. Sheet roofing's. **Form work, Scaffolding:** form work- types of formwork, centering- scaffolding- types of scaffolding. Trusses: Types- King post and queen post trusses and their uses. Stair cases: Various types of stair cases- dog legged, quarter landing, spiral stairs etc.

Text Books:

1. Engineering Materials by Rangawala. Charotar Publications
2. Building Materials by S.K.Duggal, New Age International Publications.
3. Building Materials by B.C. Punmia, Laxmi Publications private ltd.
4. Building Construction by B.C. Punmia, Laxmi Publications (p) ltd.

Reference Books:

1. Building Materials by S.S. Bhavikatti, Vices publications House private ltd.
2. Building Construction by S.S. Bhavikatti, Vices publications House private ltd.
3. Building Materials by M.L.Gambhir, Tata McGraw Hill Publishing Co. Ltd. New Delhi.
4. Building Materials by P.C.Vergheze, PHI learning (P) ltd.
5. Building construction by P.C.Vergheze, PHI Learning (P) Ltd.

CIRCUIT THEORY (For EEE)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study the concepts of passive elements, types of sources and various network reduction techniques.
2. To understand the applications of network topology to electrical circuits.
3. To study the concept of magnetic coupled circuits.
4. To understand the behavior of RLC networks for sinusoidal excitations and to understand the concept of resonance.
5. To understand the concepts of 3-phase circuits.

Course Outcomes:

Students are able to solve

1. Various electrical networks in presence of active and passive elements.
2. Electrical networks with network topology concepts.
3. Magnetic circuit with various dot conventions.
4. R, L, C network with sinusoidal excitation.
5. Three phase AC circuits.

SYLLABUS

UNIT-I: Introduction to Electrical Circuits:

Passive components and their V-I relations. Sources (dependent and independent) -Kirchhoff's laws, Network reduction techniques (series, parallel, series - parallel, star-to-delta and delta-to-star transformation). Source transformation technique, nodal analysis and mesh analysis.

UNIT-II: Network topology:

Definitions of Graph and Tree, Basic cutset and tieset matrices for planar networks, Loop and nodal methods of analysis of networks with dependent and independent voltage and current sources, Duality and Dual networks.

UNIT-III: Magnetic Circuits:

Basic definition of MMF, flux and reluctance. Analogy between electrical and magnetic circuits. Faraday's laws of electromagnetic induction Concept of self and mutual inductance. Dot convention-coefficient of coupling and composite magnetic circuit. Analysis of series and parallel magnetic circuits.

UNIT-IV: Single Phase A.C Systems:

Periodic waveforms (determination of rms, average value and form factor). Concept of phase angle and phase difference – Waveforms and phasor diagrams for lagging, leading networks. Complex and polar forms of representations, steady state analysis of R, L and C circuits. Power Factor and its significance - Real, Reactive, Apparent and Complex power. Node and mesh analysis of AC networks, Series and parallel resonance. Numerical problems.

UNIT-V: Three Phase Circuits:

Advantages of Three Phase Circuits, Balanced and Unbalanced systems, Relation between Line and Phase Quantities in Star and delta connected circuits, Analysis of Balanced & Unbalanced Three Phase Circuits.

Text Books:

1. Engineering Circuit Analysis By W.H. Hayt Jr ,J.E. Kemmerly,S.M.Durbin
8th Ed.,Mc.Graw Hill.
2. Network Analysis: Van Valkenburg; Prentice-Hall of India Private Ltd
3. Basic Electrical Engineering by V.K Mehta & Rohit Mehta- 2006 Edition
S.Chand & Company

Reference Books:

1. Circuit Theory Analysis and Synthesis., 7th Edition 2014 by Abhijit Chakrabarthi,Dhanpat Rai&Co.
2. Fundamentals of Electric circuits by Charles k Alexander & M N O Sadiku 5thEd., Mc.Graw Hill.
3. Circuits & Networks by Sudhakar & Syam Mohan, 4th edition, 2010 Mcgraw Hill.

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(For ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn the basic principles of electrical and magnetic circuits and analyze the networks
2. To understand the principle of operation and construction details of DC machines.
3. To understand the principle of operation and construction details of Transformers.
4. To understand the principle of operation and construction details of three phase Induction motor and Alternators.
5. To study the operation of PN junction diode, Half -wave Bridge rectifiers, PNP, NPN Transistors and various Amplifiers.

Course Outcomes:

1. Able to analyze the various Electrical networks and understand the basics of Magnetic Circuits
2. Able to understand the operation of DC generators,3-Point starter and conduct the Swinburne's test.
3. Able to analyze the Performance of Transformers.
4. Able to explain the operation of three phase induction motors and alternator.
5. Able to analyze the operation of Half-wave and Full-wave rectifiers and single stage CE amplifier.

SYLLABUS**UNIT I: Electrical and Magnetic Circuits:**

Basic definitions, Types of network elements, Ohm's Law, Kirchhoff's Laws, Series and parallel Circuits and star-delta and delta-star transformations-simple problems. Magnetic flux, MMF, Reluctance, Faraday's laws, Lenz's law, statically induced EMF, dynamically induced EMF.

UNIT-II: DC Machines:

Principle of operation of DC generator- EMF equation-Types of DC Generators-DC motor Types-Torque equation-Applications-Swinburne's Test, Speed control methods.

UNIT-III: Transformers:

Principle of operation of Single phase Transformers- EMF equation-losses-OC and SC Tests-Efficiency and Regulation.

UNIT-IV: AC Machines:

Principle of operation of Three phase Induction motor-Slip-Torque characteristics-Efficiency-applications- Principle of operation of Alternator-EMF equation, Regulation of alternator by synchronous Impedance method.

UNIT-V: Diodes-Rectifiers and Transistors:

PN junction diode-Forward bias and reverse bias operation, V-I characteristics-Diode applications (Half wave, Full wave and bridge rectifier), Zener diode.

PNP and NPN junction Transistors, Transistor as an amplifier, single stage CE amplifier, Frequency response of CE amplifier.

Text Books:

1. Electrical Technology by Surinder Pal Bali, Pearson Publications. January 1, 2013
2. Electronic Devices and Circuits, R.L Boylestad and Louis Nashelsky, 9th Edition, PEI/PHI 2006.
3. Principles of Electrical Engineering and Electronics, V.K mehta, Published December 1st 2006 by S. Chand & Company Ltd.
4. Principles of Electronics Engineering, V.K mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2012

Reference Books:

1. Basic Electrical Engineering, by J.B Gupta, K Kataria and Sons; Reprint 2013 edition.
2. Fundamentals of Electrical Engineering and Electronics, by Theraja B.L, S Chand Multicolor edition (1 December 2006)

**ENGINEERING PHYSICS LAB
(Common to CIV, EEE & ME)**

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

Training the field oriented Engineering graduates to handle instruments and their design methods to improve the accuracy of measurements.

1. To impart hands-on experience to the students entering engineering/Technology education about handling sophisticated equipment/ instruments.
2. To make the students understand the theoretical aspects of various phenomena experimentally.

Course Outcomes:

Physics lab curriculum gives fundamental understanding of design of an instrument with targeted accuracy for physical measurements.

1. Students get hands on experience in setting up experiments and using the instruments/equipment individually.
2. Get introduced to using new/ advanced technologies and understand their significance.

**LIST OF EXPERIMENTS
(Any 10 of the following listed experiments)**

1. Determination of wavelength of a source-Diffraction Grating-Normal incidence.
2. Newton's rings – Radius of Curvature of Plano - Convex Lens.
3. Determination of thickness of a spacer using wedge film and parallel interference fringes.
4. Determination of Rigidity modulus of a material- Torsional Pendulum.
5. Determination of Acceleration due to Gravity and Radius of Gyration-Compound Pendulum.
6. Melde's experiment – Transverse and Longitudinal modes.
7. Verification of laws of vibrations in stretched strings – Sonometer.
8. Determination of velocity of sound – Volume Resonator.
9. L- C- R Series Resonance Circuit.
10. Study of I/V Characteristics of Semiconductor diode.
11. I/V characteristics of Zener diode.
12. Characteristics of Thermistor – Temperature Coefficients.
13. Magnetic field along the axis of a current carrying coil – Stewart and Gee's apparatus.
14. Energy Band gap of a Semiconductor p - n junction.
15. Hall Effect in semiconductors.
16. Time constant of CR circuit.
17. Determination of wavelength of laser source using diffraction grating.
18. Determination of Young's modulus by method of single cantilever oscillations.
19. Determination of lattice constant – lattice dimensions kit.
20. Determination of Planck's constant using photocell.
21. Determination of surface tension of liquid by capillary rise method.

Reference Book:

1. Advanced Practical Physics –Vol 1 & 2 By S.Venkata Raman, S.Chand & Co.

ENGLISH COMMUNICATION SKILLS LAB- II
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn demonstratively the communication skills of listening, speaking, reading and writing.
2. To enable the students participate in group interactions.
3. To improve the presentation skills of the students.
4. To help the students gain their confidence in attending the interviews.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students enhance their presentation skills.
3. Students participate in group discussions and improve their team skills.
4. Students confidently face the interviews.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. **'INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

C PROGRAMMING LAB
(Common to CIV, EEE & ME)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

List of Programs

Exercise - 1 Basics

- a) What is an OS Command, Familiarization of Editors - vi, Emacs
- b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
- c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers
From Command line

Exercise - 2 Basic Math

- a) Write a C Program to Simulate 3 Laws at Motion
- b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

- a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
- b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

- a) Write a C Program to Find Whether the Given Number is
 - i) Prime Number
 - ii) Armstrong Number
- b) Write a C program to print Floyd Triangle
- c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

- a) Write a C Program demonstrating of parameter passing in Functions and returning values.
- b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III)

- a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch...case
- b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

Write a C Program to compute the values of $\sin x$ and $\cos x$ and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

- a) Search-Linear.
- b) Sorting-Bubble, Selection.
- c) Operations on Matrix.

Exercises - 9 Structures

- a) Write a C Program to Store Information of a Movie Using Structure
- b) Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
- c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

- a) Write a C Program to Access Elements of an Array Using Pointer
- b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

- a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.
- b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

Understand the difference between the above two programs

Exercise – 12 Strings

- a) Implementation of string manipulation operations **with** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare
- b) Implementation of string manipulation operations **without** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare

Exercise -13 Files

- a) Write a C programming code to open a file and to print its contents onscreen.
- b) Write a C program to copy files

Exercise - 14 Files Continued

- a) Write a C program merges two files and stores their contents in another file.
- b) Write a C program to delete a file.

Note:

- a) All the Programs must be executed in the Linux Environment. (Mandatory)
- b) The Lab record must be a print of the LATEX (.tex) Format.

ENGINEERING PHYSICS - VIRTUAL LABS – ASSIGNMENTS
(Common to CIV, EEE & ME)

Lab : 2 Periods

Int.Marks: 5

Course Objective: Training Engineering students to prepare a technical document and improving their writing skills.

Course Outcome: Physics Virtual laboratory curriculum in the form of assignment ensures an engineering graduate to prepare a /technical/mini-project/ experimental report with scientific temper.

List of Experiments

1. Hall Effect
2. Crystal Structure
3. Hysteresis
4. Brewster's angle
5. Magnetic Levitation / SQUID
6. Numerical Aperture of Optical fiber
7. Photoelectric Effect
8. Simple Harmonic Motion
9. Damped Harmonic Motion
10. LASER – Beam Divergence and Spot size
11. B-H curve
12. Michelson's interferometer
13. Black body radiation

URL: www.vlab.co.in

(Note: Internal Marks of **Engineering Physics - Virtual Labs – Assignments** are to be considered as Assignment marks in the Internal Marks of **Engineering Physics- B17 BS 1204**)

Code: B17 BS 1211

NCC
(Common to CIV, EEE & ME)

Practice : 2 Periods

The NCC- National Integration and Awareness- Drill- Personality Development Life Skills- Leadership- Disaster Management-Social Awareness and Community Development- Health and Hygiene- Environment Awareness and Conservation.

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

II/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

CIVIL ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 2101	Mathematics -IV	3	3	1	--	4	30	70	100
B17 CE 2101	Mechanics of Solids	3	3	1	--	4	30	70	100
B17 CE 2102	Environmental Engineering - I	3	3	1	--	4	30	70	100
B17 CE 2103	Building Planning & Design	3	2	--	3	5	30	70	100
B17 CE 2104	Surveying-I	3	3	1	--	4	30	70	100
B17 CE 2105	Engineering Geology	3	3	1	--	4	30	70	100
B17 CE 2106	Engineering Geology Lab	2	--	--	3	3	50	50	100
B17 CE 2107	Strength of Materials Laboratory	2	--	--	3	3	50	50	100
B17 CE 2108	AutoCAD for Civil Engineering	1	--	--	2	2	50	--	50
B17 BS 2107	English Proficiency-I	--	1	1	--	2	--	--	--
B17 BS 2108	Professional Ethics & Human Values	--	2	--	--	2	--	--	--
Total		23	20	6	11	37	330	520	850

MATHEMATICS IV
(Common to CE,ECE,EEE& ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students should learn

1. The concept of Analytic function, its implications and applications in flow problems.
2. Solution of one- dimensional wave equation, one-dimensional heat equation and two-dimensional Laplace equation by the use of 'separation of variables'.
3. Formation and solution of linear difference equations. Important concepts of Z-transform and its use to solve linear difference equations.
4. Basic concepts of certain discrete and continuous probability distributions.
5. Concepts of Sampling theory for analyzing large and small samples.

Course Outcomes:

Students will be capable of

1. Using the concept of Analytic function in applications including Electrostatics and Fluid dynamics.
2. Finding theoretical solution of certain Elliptic, Parabolic and Hyperbolic partial differential equations.
3. Using Z-transforms to solve linear difference equations with constant coefficients.
4. Fitting of probability frequency distribution to a given data.
5. Using the concepts of sampling theory to analyze data related to some large and small samples.

SYLLABUS

UNIT-I Functions of a Complex Variable

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n , e^z , $\sin z$, $\sinh z$ and $\log z$ (**no questions may be set**).

Limit and continuity of a function of the complex variable, derivative, analytic function, entire function, Cauchy- Riemann equations, finding an analytic function, Milne-Thomson method, Applications of analytic function to flow problems, and in Electrostatics. Conformal mapping: the transformations defined by $w = z+c$, $w = cz$, $w = 1/z$, the Bilinear transformation, $w = z^2$ and $w = e^z$.

UNIT-II Applications of Partial Differential Equations

Method of separation of variables, One –dimensional wave equation, the D'Alembert's solution, one-dimensional heat equation, two-dimensional heat flow in steady state (solution of two-dimensional Laplace equation in Cartesian coordinates only)

UNIT-III Difference Equations And Z-Transforms

Formation of a difference equation, Rules for finding complimentary function and particular integral for linear difference equations.

Definition of Z- transform, some standard Z- transforms, properties, transform of a function multiplied by n , initial value theorem and final value theorem(without proof), evaluation of inverse Z- transforms, convolution theorem (without proof), solution of linear difference equations by the use of Z- transforms.

UNIT-IV Probability Distributions

Binomial distribution, Poisson distribution, Normal distribution: Definition (pmf/pdf), notation, mean, variance, moment generating function, probability generating function and fitting of a distribution.

UNIT-V Sampling Theory

Sampling theory: Sampling distribution, standard error, testing of Hypothesis, level of significance, confidence limits, simple sampling of attributes, sampling of variables, estimation of mean and variance.

Large samples: testing of hypothesis for sample proportion, two proportions, single mean and two means.

Small samples: Degrees of freedom, Students' t- distribution, t-test for single mean, two means; Chi-squared distribution-testing the goodness of a fit.

Text Book:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National Publishing Company.

MECHANICS OF SOLIDS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To introduce the concepts of stress, strain and elastic constants and their relations.
2. To familiarize with shear force, bending moment and torsion induced and shear stresses and bending stresses developed for different sections of beams and shafts.
3. To impart the knowledge on the concept of principal stresses and principal strains.
4. To familiarize with stresses and strains induced in columns.

Course Outcomes: At the end of the Course, Student will be able to:

1. Summarize the behavior of basic materials under the influence of different external loading conditions and support conditions.(K2)
2. Determine shear Force and Bending moments in statically determinate Beams and draw the Diagrams.(K5)
3. Examine the different methods to find slope and deflection of beams subjected to loads(K4)
4. Estimate the principal stresses & strains and torsional stresses in structural members(K3)
5. Evaluate the crippling load for columns with different end conditions (K5)

SYLLABUS

UNIT-I Simple stresses & Strains:

Definitions of stress and strain – types of stresses and strains – Elasticity – Hooke's law – Stress – Strain diagram for Mild steel – working stress- factor of safety- Lateral strain – Poisson's ratio and volumetric strain – Elastic Moduli and the relationship between them – Bars of varying section – composite bars – temperature stresses.

Strain Energy

Definition – Resilience – SE due to gradually applied; suddenly applied and impact loads – simple applications.

UNIT-II Shear Force & Bending Moment Diagrams:

Definition of beam – Types of beams – concept of SF and BM – SF & BM diagrams for cantilever, SS and overhanging beams subjected point loads, UDL, Uniformly varying loads and combination of these loads – point of contraflexure – Relationship b/w S.F, BM and rate of loading.

UNIT-III Flexural Stresses:

Theory of simple Bending – Assumptions–Derivation of Bending equation - ($\frac{M}{I} = \frac{F}{Y} = \frac{E}{R}$)

Neutral axis – Determination of bending stresses – section modulus of rectangular, & Circular sections (Solid and Hollow), I,T, channel sections – Design of simple beam sections.

Shear Stresses

Derivation of shear stress formula – shear stress distribution across various beam sections like rectangular, circular, Triangular, I, T, angle sections, built up beams, Definition of shear centre.

Deflections of Beams: (i) Cantilever (ii) Simply supported and (iii) Over hanging beams using (a) Double integration and (b) Macaulay's method.

UNIT-IV Principal Stresses and strains:

Introduction-stresses on an inclined section of a bar under axial loading- compound stresses-Normal and tangential stresses on an inclined plane for biaxial stresses-Two perpendicular normal stresses accompanied by a state of simple shear-Mohr's circle of stress-Principal planes and principal stresses- Construction of Mohr's Circle (graphical Method)

Torsion of Circular Shafts

Theory of pure Torsion – Derivation of Torsion equation $\left(\frac{T}{J} = \frac{\tau_{max}}{R} = \frac{G\theta}{l}\right)$ – Torsional moment of Resistance – polar section Modulus – power transmitted by a shaft – combined bending and torsion.

Springs

Types of springs – springs in series and parallel – close coiled helical springs

UNIT-V Columns & Struts:

Introduction – short, medium and long columns – axially loaded compression members – crushing load – Buckling load (or) critical load (or) crippling load – Euler's theory for long columns – Assumptions – Derivations of Euler's critical load formula for various end conditions – Effective length of column – slenderness ratio – limitations of Euler's Theory – Rankine formula – for both long and short columns – column subjected to Eccentric loading – Euler's Method and prof. Perry's formula.

Text Books:

1. Ramamrutham ,Strength of materials, Dhanpat rai publishers.
2. Vazrani and Ratwani ,Strength of materials, Khanna Publishers .
3. T.D.Gunneswara Rao and Andal Mudimby,Strength of Materials Fundamentals and Applications, Cambridge University Press.

Reference Books:

1. Timoshenko and Young, Elements of strength of materials, East West press private Ltd.
2. Popov, Introduction to mechanics of solids , Englewood cliffs N.J Prentice Hall.
3. DrR.K.Bansal, Strength of materials , Laxmi Publications .
4. S.S.Bhavikatti, Strength of materials, Vikas Publications.

ENVIRONMENTAL ENGINEERING - I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Student shall be able to:

1. Explain the fundamentals of chemistry involved in water
2. Understand the analysis of water quality parameters
3. Understand the working principles of various unit operations of water treatment plant
4. Understand the importance of modern membrane techniques employed for water treatment
5. Design the water treatment unit operations.

Course Outcomes:

By the end of the course, the student should be able to:

1. Explain the quality of water[K2]
2. Analyze the water quality parameters and compare with the permissible limits [K4]
3. Summarize the working principles of conventional unit operations of a water treatment plant [K2]
4. Determine the sizes of different unit operations in a water treatment plant [K5]
5. Assess the suitability of conventional methods and latest membrane processes for different water bodies [K5]
6. Design a conventional water treatment plant with given specifications for given capacity [K6].

SYLLABUS**UNIT – I**

Necessity and objectives of protected water supply system - Flow chart of public water supply system – Role of environmental agencies

Water Demand and Quantity studies: Estimation of water demand for a town or city, Types of water demands, Per capita Demand, Factors affecting the Per Capita Demand, Variations in the Demand, Design Period, Factors affecting the Design period, Population Studies, Population Forecasting Studies.

UNIT - II

Hydrological Concepts: Hydrological Cycle - Types of Precipitation - Measurement of Rainfall. Surface sources of water: Lakes, Rivers, Impounding Reservoirs - Capacity of storage reservoirs - Mass curve analysis. Groundwater sources of water: Types of water bearing formations – Springs - Wells and Infiltration galleries, Yields from wells and infiltration galleries.

UNIT – III

Collection of Water: Factors governing the selection of the intake structure - Types of Intakes - Conveyance of Water: Gravity and Pressure conduits - Types of Pipes - Pipe Materials, Pipe joints - Design aspects of pipe lines - Laying of pipe lines.

Quality and Analysis of Water: Characteristics of water – Physical, Chemical and Biological. Analysis of Water – Physical, Chemical and Biological - Impurities in water - Water borne diseases - Drinking water quality standards.

UNIT – IV

Treatment of Water: Flowchart of water treatment plant, Treatment methods (Theory and Design) - Sedimentation, Coagulation, Sedimentation with Coagulation, Filtration, Chlorination and other Disinfection methods.

UNIT – V

Softening of Water, Defluoridation, Removal of odours.

Distribution of Water: Methods of Distribution system - Components of Distribution system .

Text Books:

1. S.K.Garg, Water Supply engineering, 17th Edition, Khanna Publishers.
2. K.N. Duggal, Elements of Environmental Engineering, McGraw Hill Publishers

Reference Books:

1. Peavy, Rowe, Tchobanoglous, Environmental Engineering, McGraw Hill Publishers
2. S.C.Rangwala, Water Supply and Sanitary Engineering, Charotar Publishers.

BUILDING PLANNING & DESIGN

Lecture	: 2Periods	Int.Marks	: 30
Lab	: 3 Periods.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand the principles of planning and byelaws.
2. To draw plan, elevation and section of load bearing and framed structures.
3. To prepare detailed drawings for doors, windows, etc.

Course Outcomes:

1. Understand various types of buildings and housing concept.
2. Apply the concepts of climatology and orientation of both residential and commercial buildings.
3. Apply the principles of planning and byelaws used for building planning.
4. Recommend appropriate planning for 2 Bed room and 3 Bed room houses.
5. Draw plan, elevation and section for various structures.
6. Design individual rooms with attention to functional and furniture requirements.

SYLLABUS**UNIT – I Climatology:**

Elements of climate: Sun, Wind, Relative Humidity, and Temperature. Comfort conditions for house. Various types of Macroclimatic zones, Design of Houses and layouts with reference to climatic zones, Ventilation, Principles of Planning, Orientation of Buildings.

UNIT – II Residential Buildings:

Different types of Residential Buildings, Selection of site for residential buildings. Guidelines for planning and drawing of residential building. General building regulations and byelaws for residential buildings.

UNIT-III Preliminary Drawings :

(a) Conventional signs of materials various equipment used in a Residential Building (copying exercise) (b) Plan section and Elevation of a small House (one room and verandah) (copying exercise) (c) Plan section and Elevation of Two Bed Room House (copying exercise) (d) Plan section and Elevation of three bed room house in Hot and Humid zone, Hot and Arid zone, cold zone (copying exercises) (e) Design of Individual rooms with particular attention to functional and furniture requirements. Building regulations and Byelaws of Residential Buildings . Drawing the Plan Section and Elevation of Houses with given Functional requirements and climatic data. (Emphasis may be given to Hot and Humid zones.)

Note:

The question paper consists of Part-A and Part-B. Part-A consists of 4 questions, 2 questions for each of Unit – I & II and Part-B consists of a compulsory question for 38 marks

Text Books:

1. Building Planning and Drawing by Dr.N. Kumara Swamy and A.Kameswara Rao, Charotar Publishing House.
2. Building Planning Drawing and Scheduling by Gurucharansingh and Jagadish Singh, Standard Publishers Distributors.

Reference Books:

1. Building Drawing with an integrated approach to Builtenvironment by M.G.Shah, C.M.Kale and S.Y.Patki, McGraw-Hill Publishing Company Limited, NewDelhi.
2. Civil Engineering Drawing Series 'B' by R.Trimurty, M/S Premier PublishingHouse.
3. Relevant NPTELCourses.

SURVEYING-I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Student shall be able to

1. Learn and understand the basic methods of classifications and principles of surveying.
2. Learn how to differentiate the methods of area calculation used in surveying.
3. Differentiate the Angles and Bearings and able to identify direction of a line
4. Describe Plane tabling methods of Surveying
5. Construct the Leveling methods to identify Elevation of the required points
6. Differentiate the usage of Minor Instruments in surveying.

Course Outcomes:

After the completion of the course student should be able to

1. Appreciate the importance of preparation of Map and Plan for required site with suitable scale.
2. Prepare contour Map and Estimate the Quantity of earthwork required for formation level for Road and Railway Alignment.
3. Judge on which type of instrument to be used for carrying out survey for a specific work
4. Describe different modern instruments used in surveying.

SYLLABUS**UNIT-I Introduction :**

Classification and principles of surveying. Triangulation and Trilateration Earth as spheroid, datum, geoid, Azimuth, latitude, longitude, Map projections, scales, plans, & Maps. Chain surveying: Instrumentation for chaining – Errors due to incorrect chain-Chaining on uneven and sloping ground-Errors in chaining-Tape corrections – Problems: Base line measurement- chain Triangulation-Check lines, Tie lines, Offsets. Basic problems in chaining, obstacles in chaining-Problems-Conventional signs.

UNIT -II Compass Survey :

Introduction to compass survey Definitions of Bearing. True bearing, True meridian, Magnetic Meridian, Magnetic bearing – Arbitrary Meridian, R.B. & B.B of lines – Designation of bearings – W.C.B. & R.B. – Conversion of bearings from one system to the other Related problems – Calculation of angles for bearings, Calculation of bearing for angles, Related problems – Theory of Magnetic compass (i.e. Prismatic compass) – Magnetic dip-Description of Prismatic compass. Temporary adjustments of compass-Magnetic Declination – Local attraction-Related Problems-Errors in compass survey.

UNIT- III Traverse Surveying :

Chain and compass traversing-Free or loose needle method – Fastneedle method-Checks in closed and open traverse-Plotting methods of traverse Survey - Closing error-Balancing the traverse-Bowditch's method-Transit method, Gale's Traverse tablePlane table surveying:

Introduction-Advantages, Accessories-Working operations such as fixing the table to tripod, levelling-centering-orientation by back-sighting.

UNIT -IV Levelling :

Definitions of terms-Methods of levelling-Uses and adjustments of dumpy level-Temporary and permanent adjustments of dumpy level levelling staves-Differential leveling, Profile levelling-Cross sections-Reciprocal levelling. Precise levelling-Definition of BS, IS, FS, HI, TP-Booking and reduction of levels, H.I. methods-Rise and fall method-Checks-Related problems-Curvature and Refraction Related Problems-Correction-Reciprocal levelling- Related problems-L.S & C.S Levelling-Problems in levelling-Errors in levelling.

Contouring:

Definitions-Interval, Characteristics of contours-methods of locating contours. Direct and indirect methods-interpolation of contours-Contour gradient-Uses of contour maps. Contours mapping using computer techniques (surfer, CAD)

UNIT -V: Minor instruments :

Uses and adjustments of the following minor instruments:

Line Ranger, Optical Square, Abney level, Clinometer, Ceylon Ghattracer, Pantagraph, Sextant and Planimeter.

Text Books:

1. Surveying By Dr. K.R. Arora, Standard Book House.
2. Surveying Vol. 1 and 2 By Punmia, Standard Book House.

Reference Books:

1. Surveying Vol. 1 and 2 – By S.K. Duggal, Tata Mc.Graw Hill Publishing Co.
2. Surveying: Theory & Practices by James M. Anderson and Edward M. Mikhail

ENGINEERING GEOLOGY

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

To introduce basic geology and the principles of site investigation to civil engineering students that includes:

1. Sedimentary processes (Weathering, erosion, deposition).
2. Geological Structures (Joint, veins, crack, faults, and fold), reasons of formation for each type and their effects on the engineering projects.
3. The basics of Remote sensing and its applications in Civil Engineering.
4. Types of Mass wasting, factors increasing instability and methods of stabilization.
5. Types of Dams and Tunnels.
6. Study ground water, factors affecting and the methods of searching for ground water.

Course Outcomes: At the end of the Course,

1. The course will provide the students with basic knowledge and understanding in the most central part of engineering geology, rock and soil.
2. Students should develop an appreciation of geologic processes and their influence civil engineering works.
3. The course will give students an overview and an understanding of the engineering properties of rock and soil materials.
4. Based on lectures and exercises, students will gain basic understanding of the importance of engineering geology related to technical issues during construction.
5. Students will develop the ability to perform basic engineering geological assessments and analysis, and to understand the relevance of engineering geology in complex projects in and on solid rock.

SYLLABUS**UNIT-I Introduction to General Geology:**

Importance of geology from civil engineering point of view. Branches of Geology. Weathering – types and its engineering importance; Erosion, Soils: Soil profile, soil formation, types of Indian soils. Land forms produced by: running water and glaciers, Land Wind, Sea Waves and Currents and Ground Water.

UNIT-II Mineralogy & Petrology:

Mineralogy: Mineral definition, physical properties of minerals. Study of important rock forming minerals: Silicate structures, Quartz, Feldspars, Pyroxenes, Amphiboles, Micas and Clays.

Petrology: Definition of rock. Types of rocks - Igneous rocks: Granite, Syenite, Dolerite, Gabro, Diorite, Basalt. Sedimentary rocks: Breccia, Conglomerate, Sandstone, Shale, Limestone. Metamorphic rocks: Gneiss, Khondalite, Schist, Slate, Marble, Quartzite, Charnokite. Engineering properties of rocks.

UNIT-III Statigraphy& Structural Geology:

Statigraphy: Geological Time scale, Major geological formations of India and their geological importance - Achaeans, Cuddapahs, Vindhyans, Gondwanas and Deccan Traps. Mineral resources of Andhra Pradesh.

Structural Geology: Elements of Structural Geology - Strike, Dip, Plunge. Working principles of Clinometer compass and Brunton Compass and their use in Civil Engineering. Study of Geological Structures - Folds, Faults and Joints.

UNIT-IV Remote Sensing and Geophysical Methods:

Remote Sensing: Introduction, Electromagnetic Spectrum, Aerial Photographs: types of aerial photos. Elements of photo interpretation. Satellite Remote Sensing: Satellites, Sensors and Data Products. Principles of Geographical Information Systems. RS and GIS applications to Civil Engineering.

Geophysical Methods: Principles of Geophysical Methods, Electrical, Seismic, Gravity and Magnetic. Principle of Resistivity method and configurations. Applications of Resistivity Method: Soil Profile, Hard rock and Ground Water Table. Principles of Seismic refraction and reflections methods and their applications to Civil Engineering problems.

UNIT-V Geological Investigations:

Role of Engineering Geologist in planning, design and construction and post construction stages in Civil Engineering works. Geological investigations for Dams and Reservoirs and Tunnels, Geological investigations for bridges and multistoried structures. Geological investigations for highways, air fields and railway lines. Geological investigations for Coastal structures and Environmental Geology.

Text Books:

1. Engineering and General Geology by Parbin Singh – Katson Publishing House
2. Engineering Geology by N.Chennakesavulu, Mc-Millan, India Ltd. 2005
3. Principles of Engineering Geology by KVGK Gokhale. B.S Publications-2005
4. Principles and Applications of Photo Geology by Pandey, Willey Eastern Limited
5. Engineering Geology by K.M.Bangar

Reference Books:

1. Engineering Geology and Geo-techniques by F.G. Bell
2. Fundamentals of Remote Sensing by George Joseph. University Press (India) Private Limited.
3. Fundamentals of Engineering Geology by F.G. Bell, Button Wortus Landon

ENGINEERING GEOLOGY LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int. Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To identify the formation of minerals
2. To understand the mega-scopic identification of rocks and minerals
3. To understand the importance of geophysical methodologies
4. To understand the geological maps

Course outcomes:

After completion of the lab, Students will be able to

1. Elucidate the mega-scopic identification of minerals
2. Categorize the rocks according to mega-scopic description
3. Interpret geological maps
4. Estimate the types of subsurface formation by using geophysical methods

List of Experiments:

1. Study of physical properties and identification of minerals.
2. Megascopic identification of rocks and their Engineering properties.
3. Description and Identification of Geomorphological models.
4. Description and Identification of Structural models.
5. Simple Structural Geology problems.

Lab Examination Pattern:

1. Description and identification of SIX minerals.
2. Description and identification of SIX rocks (Igneous, Sedimentary and Metamorphic rocks).
3. Identification of Geomorphology model.
4. Identification of Structural geology model.
5. Problem on Strike and Dip.

Reference Books:

1. Engineering Geology and Geo-techniques by F.G. Bell
2. Fundamentals of Remote Sensing by George Joseph. University Press (India) Private Limited.
3. Fundamentals of Engineering Geology by F.G. Bell, Button Wortus Landon

STRENGTH OF MATERIALS LABORATORY**Lab : 3 Periods****Exam : 3 Hrs.****Int.Marks : 50****Ext. Marks : 50****Credits : 2****Course Objectives:** Student will be able to

1. The student will be able to study the uses of various types of testing machines and capacities.
2. The student will be able to test the structural materials against various straining actions which the member is intended to bear during its service. The strength of materials against various straining actions can be determined by conducting the appropriate tests.
3. The student will be able to identify, understand or design various implements for generating required straining forces using the machine or to augment the utility of the same machine.
4. The student will be able to understand the principle of self- straining systems and to analyze the various elements of the machine and appreciate the capacity of the testing machine or the frame.

Course Outcomes: On successful completion of this course, it is expected that students should be able to

1. The student clearly understands the concepts of deciding the shape or type of specimen for assessing the respective strengths against various straining actions.
2. The student can design the specimens for assessing a particular property of the material with the available machines.
3. The student can design the experiments making use of various techniques of load measuring or deformation measuring instruments.
4. The student will be confident to decide the range of the machine and set the machine accordingly by suitable modifications, for results with a finer degree of accuracy.

List of Experiments:

1. Tension test on Mild/HYSD Steel bar
2. Compression test on wood (Parallel and Perpendicular to grains)
3. Test on spring
4. Brinell's Hardness test
5. Charpy and Izod impact tests
6. Double Shear test
7. Bending test on Steel and Wood as Simply supported beams and Cantilever beams.
8. Verification of Maxwell's Reciprocal theorem on simply supported beam.

Reference Books:

1. P.N. Singer and P.K. Jha , Elementary Mechanics of solids , New Age International Pvt .Ltd

AUTOCAD FOR CIVIL ENGINEERING

Lab : 2 Periods

Exam : 3 Hrs.

Int. Marks : 50

Credits : 1

Course Objectives:

1. Increase ability to communicate with the people.
2. Learn to sketch and take field dimensions.
3. Learn basic AutoCAD skills.
4. Learn basic engineering drawing formats.
5. Prepare the student for future Engineering positions.

Course Outcomes:

1. Student's ability to perform basic sketching techniques will improve.
2. Student's ability to use architectural and engineering scales will increase.
3. Student's ability to produce engineered drawings will improve.
4. Student's ability to convert sketches to engineered drawings will increase.
5. Student's will become familiar with office practice and standards.
6. Student's will become familiar with AutoCAD two dimensional drawings.
7. Student's will develop good communication skills and teamwork.

SYLLABUS

Fundamentals of Computers

1. Introduction
2. Computer Hardware and Software Concepts
3. Introduction of Personal Computer and Operating Systems WINDOWS-XP, Windows- 7, File Management
4. Drawing using AutoCAD
5. Starting a New Drawing/Opening an existing drawing
6. Drawing Commands
7. Hatching Command Text (multi-line & single line) and Formatting Text Styles
8. View Commands & Drawing Settings and Aids
9. Modify Commands
10. Dimension Command Formatting Dimension Style and Multi-leader Style
11. Drawing Settings and Aids
12. Saving and Plot
13. Simple Building Drawing

Reference Books:

1. Learning Auto CAD 2010 Volume-I , Autodesk.
2. Auto CAD 2013 fundamentals- Elsenloss, SDC Publications

ENGLISH PROFICIENCY-I
(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

The Students will be able to

1. Communicate their ideas and views effectively.
2. Practice language skills and improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self attributes which require improvement and motivation.
5. Build confidence and overcome their inhibitions, stage freight, nervousness, etc.
6. Improve their innovative practices in speaking.

Course Outcomes:

The Students will

1. Improve speaking skills.
2. Enhance their listening capabilities.
3. Learn and practice the skills of composition writing.
4. Enhance their reading and understanding of different texts.
5. Improve their inter-personal communication skills.
6. Be confident in presentation skills.

SYLLABUS

UNIT-I :LISTENING

Selected Motivational Speeches

Selected Moral Stories

UNIT-II :SPEAKING

Book Review

Skit Presentation

PowerPoint Presentations

Describing event/place/thing

Extempore

Group Discussion

Picture Perception and Describing Test

UNIT-III : READING

Speeded Reading

Reading Comprehension

UNIT-IV :WRITING

Paragraph Writing

Literary Appreciation – Understanding the Language of Literature

UNIT-V :PROJECT

Ad Making

Reference Books:

1. Classic Short Stories-A Reader's Digest Selection
2. English for Colleges by Brendan J. Carroll, Macmillan Publications
3. The World's Great Speeches- edited by B.S. Sekhar, Jeet Publications
4. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
5. English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
6. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
7. The Oxford Guide to Writing and Speaking by John Seely OUP

PROFESSIONAL ETHICS & HUMAN VALUES
(Common to CIVIL,EEE & MECH)

Lecture : 2 Periods.
Exam : ---

Int.Marks : --
Credits : --

Course Objectives:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

Course outcomes:

By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS**UNIT – I**

Ethics and Human Values: Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

UNIT – II

Engineering Ethics: Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics

Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

UNIT – III**Engineering as Social Experimentation:**

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders Role of engineers in promoting ethical climate, balanced outlook on law.

UNIT – IV**Safety Social Responsibility and Rights:**

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

UNIT – V**Global Issues:**

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

Text Books:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

References:

1. Charles D, Fleddermann, "Engineering Ethics", Pearson / PHI, New Jersey 2004 (Indian Reprint).

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

II/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)

CIVIL ENGINEERING
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 CE 2201	Structural Analysis -I	3	3	1	--	4	30	70	100
B17 CE 2202	Fluid Mechanics- I	3	3	1	--	4	30	70	100
B17 CE 2203	Environmental Engineering - II	3	3	1	--	4	30	70	100
B17 CE 2204	Concrete Technology	3	3	1	--	4	30	70	100
B17 CE 2205	Surveying - II	3	3	1	--	4	30	70	100
B17 CE 2206	Remote Sensing & GIS	3	3	1	--	4	30	70	100
B17 CE 2207	Surveying Field Work	2	--	--	3	3	50	50	100
B17 CE 2208	Fluid Mechanics Lab- I	2	--	--	3	3	50	50	100
B17 CE 2209	Industry Oriented Technology Lab	1	--	--	2	2	50	--	50
B17BS 2206	English Proficiency-II	--	1	1	--	2	--	--	--
Total		23	19	7	8	34	330	520	850

STRUCTURAL ANALYSIS - I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To familiarize with the deflection of simple determinate beams
2. To familiarize with stresses and strains induced in Thin & Thick cylinders
3. To impart knowledge on various energy methods and theories of failure
4. To analyze the propped cantilever and fixed beams
5. To determine the deflections of determinate beams using energy methods
6. To draw the influence lines and moving loads for statically determinate beams
7. To analyze the statically indeterminate structures using energy methods

Course Outcomes:

Student will be able to

1. Determine deflections in determinate beams by different methods
2. Evaluate the strain energy for structural members subjected to different loads
3. Analyze different indeterminate beams for BM and SF by different methods of analysis
4. Determine reactions, BM & SF in beams subjected to moving loads
5. Distinguish between thin and thick cylinders and understand different failure theories

SYLLABUS**UNIT – I: Deflections of Beams**

By using (i) Moment area method (ii) Conjugate beam method (iii) Unit load method (iv) Castigliano's theorem-1.

Strain- energy due to (i) Axial load (ii) Bending Moment (iii) Shear force and (iv) Torque

Deflections of Statically Determinate Structures: (a) Single storey, single bay rectangular portal frames using (i) Unit load method, (ii) Castigliano's theorem-1. (b) Trusses (having 9 members or less) using (i) Unit load method, (ii) Castigliano's theorem-1.

UNIT – II: Propped Cantilevers, Fixed Beams

Analysis of propped cantilever by method of consistent deformation.

Fixed Beams: Fixed end moments for beams of uniform section for different types of loading; Effect of sinking of support; effect of Rotation of a support; BMD for fixed beam.

UNIT – III: Continuous Beams

1. Analysis of continuous beams by
2. Theorem of three moments
3. Slope deflection method
4. Moment distribution method
5. Kani's method.

UNIT – IV: Influence Lines and Moving Loads

Definition – Influence line for Reaction, SF and BM-Load position for Max SF at a section –Load position for max BM at a section- Single point load, U.D.L longer than the span, U.D.L.shorter than the span- Focal length.

Introduction of moving loads – Max SF and BM at a given section and absolute Max SF and BM due to single concentrated load, U.D.L. longer than the span, U.D.L. shorter than the span, two point loads with fixed distance between them and several point loads, Maximum Bending moment at a section under a wheel load and absolute maximum Bending moment in the case of several wheel loads- Equivalent uniformly Distributed load.

UNIT -V: Thin cylinders:

Calculation of longitudinal and hoop stresses in thin cylinders subjected to internal pressure, Wire wound thin cylinders.

Thick cylinders- Lamé's theory, Compound tubes.

Theories of failures (i) Maximum Principal stress theory, (ii) Maximum Principal strain theory, (iii) Maximum shear theory (iv) Maximum strain energy theory and (v) Maximum distortion theory.

Text Books:

1. Structural Analysis Volume _II By Vazrani and Ratwani , Khanna Publishers
2. Strength of Materials – Ramamrutham, Dhanpat Rai Publishing Company
3. T.D.Gunneswara Rao and Andal Mudimby, Strength of Materials Fundamentals and Applications, Cambridge University Press.

Reference Books:

1. Elementary strength of materials – Timoshenko and Young , East west press Pvt Ltd
2. Strength of materials by Dr. R.K. Bansal ,Laxmi Publications (p) Ltd
3. Strength of materials – Volume – I by S.S.Bhavikatti ,Vikas Publishing house (p) Ltd

FLUID MECHANICS- I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To develop an insight into engineering problems related to fluids.
2. Student is expected to learn about the pressure at a point, forces on fluid element to solve complex problems in engineering.
3. Student shall be able to know different types of fluid flows and apply the principles of conservations of mass, momentum and energy.

Course Outcomes:

By the end of the course the student will be able to

1. Determine the physical properties of fluids and different types of forces acting on a fluid element extended to forces on various gates.
2. Determine the forces that are acting on immersed bodies in static fluids through application of buoyancy and floatation.
3. Determine different types of fluid flows to find out the local and convective accelerations in 1D, 2D flows fields and derive the Laplace equation.
4. Apply conservation principles of mass momentum and energy on fluids through system and control volume approaches.
5. Calculate the force exerted by the fluid on bends, nozzles, plates and vanes by impulse momentum principle.
6. Analyze the steady laminar and turbulent flows through pipes and solve pipe networks for series and parallel pipes to solve two reservoir and three reservoir problems.

SYLLABUS**UNIT – I:**

Basic Fluid Properties: Definition of Fluid, basic properties of fluid, Viscosity - Newton's Law of Viscosity, Capillarity and Surface Tension.

Fluid Pressure: Fluid Pressure at a point, Pascal's law, Variation of pressure with elevation, Hydrostatic law, Absolute, Gauge and Vacuum Pressures. Pressure measurement – Piezometers, Manometers and Pressure Gauges. Centre of Pressure, Forces on submerged surfaces, crest gates and lock gates.

UNIT – II:

Buoyancy and Floatation: Archimedes Principle- Buoyancy & Floatation - Stability of Floating Bodies- Centre of Buoyancy - Metacentric Height and its Determination.

Fluid Kinematics: Types of fluid flow, Velocity, Rate of flow, Continuity Equation, Streamline, Path line, Streak line, Local, Convective and Total Acceleration; One & Two Dimensional Flows. Stream Function, Velocity Potential- Rotational & Irrotational Flows, Laplace Equation, Flow net.

UNIT – III:

Fluid Dynamics: Energy possessed by fluid in motion, Euler's equation of motion - Bernoulli's equation. Energy correction factor.

Flow through orifices and mouth pieces: Types of orifices and mouth pieces, coefficient of contraction, velocity and discharge.

Flow through notches and weirs: Types of notches and weirs, Measurement of discharge.

UNIT – IV:

Impulse momentum equation – Momentum correction factor, Forces on pipe bends and reducers. Angular Momentum – Torque and work done; Sprinkler Problems.

Laminar Flow: Relation between shear and Pressure Gradients in Laminar Flow; Reynold's experiment; Critical velocity; Steady laminar flow through a circular pipe – Hagen Poiseuille's Law.

UNIT – V:

Flow through pipes: Flow measurement through pipes – Venturimeter, orificemeter, nozzle meter. Loss of head, head loss due to friction – Darcy –Weisbach equation, minor losses, Total Energy Line, Hydraulic Gradient Line. Pipes in Series, pipes in parallel. Problems on Two reservoir and three reservoir flows. Water hammer, surge tanks.

Text Books:

1. Fluid Mechanics and Hydraulic Machinery by P.N. Modi & S.M. Seth, Standard Book House.
2. Fluid Mechanics by A.K. Jain, Khanna Publishers

Reference Books:

1. Hydraulics Fluid Mechanics and Fluid Machines, S.Ramamrutham, Dhanpat Rai Publishing Co.
2. Engineering Fluid Mechanics by K.L. Kumar, S. Chand & Co
3. Fluid Mechanics and Hydraulic Machines by R.K. Bansal, Laxmi Publications.
4. F M White, Fluid Mechanics, Tata McGraw Hill Publication 2011.
5. Relevant NPTEL Courses.

ENVIRONEMNTAL ENGINEERING - II

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Student shall be able to:

1. Explain the fundamentals of chemistry and biology involved in sewage treatment
2. Understand the principles and working of each process that involve physical and biological reactions
3. Design the treatment unit operations

Course Outcomes:

By the end of the course, the student should be able to:

1. Compare water and waste water
2. Explain principles of conventional treatment process and miscellaneous treatment techniques
3. Examine the operational differences of each unit process
4. Interpret the feasible technique required for particular waste water
5. Determine the size of unit operations using working principles of each.
6. Design a sewage treatment plant assuming whatever data is required .

SYLLABUS**UNIT – I**

Sanitation – systems of sanitation– sewerage systems – Estimation of sewage flow and storm water – design of sewers –sewer types – Layout of sewer network – materials for sewers – cleaning and ventilation methods – sewer appurtenances

UNIT – II

Pumping – necessity – pumping station – its location – functions – types of pumps- sewage and water pumping requirements- House drainage systems – traps – classification – drain pipes – plumbing systems – Layout of building drainage – Storm water drainage

UNIT – III

Quality of sewage –Sampling and analysis - decomposition- cycles of decomposition – BOD – COD – Layout of sewage treatment plant – F/M importance – preliminary treatment – screens – grit chamber – skimming tanks – Primary treatment – Sedimentation tanks

UNIT – IV

Secondary treatment – Attached growth process – contact beds – intermittent sand filters – trickling filters – Suspended growth process – activated sludge process- Miscellaneous methods for sewage treatment – oxidation ponds - oxidation ditches - RBC's - sewage lagoons- extended aeration process – septic tanks – imhoff tanks

UNIT - V

Sewage disposal methods – natural and artificial methods –Sludge generation- characteristics - treatment- sludge disposal methods.

Text Books:

1. S.K.Garg, Sewage disposal and Air pollution Engineering, Khanna Publishers

Reference Books:

1. Peavy, H.S. Rowe and Tchobanoglous, Environmental Engineering, Mc. Graw Hill publishers
2. S.C.Rangwala, Water Supply and Sanitary Engineering, Charotar Publishers.

CONCRETE TECHNOLOGY

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn the concepts of Concrete production and its behavior in various environments.
2. To learn the test procedures for the determination of properties of concrete.
3. To understand durability properties of concrete in various environments.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand the basic concepts of concrete.
2. Realize the importance of quality of concrete.
3. Familiarize the basic ingredients of concrete and their role in the production of concrete and its behavior in the field.
4. Test the fresh concrete properties and the hardened concrete properties.
5. Evaluate the ingredients of concrete through lab test results.
6. Design the concrete mix by BIS method.
7. Familiarize the basic concepts of special concrete and their production and applications.
8. Understand the behavior of concrete in various environments.

SYLLABUS**UNIT-I :****Ingredients Of Concrete**

Cements &Admixtures: Portland cement – Chemical composition – Hydration, Setting of cement, Fineness of cement, Structure of hydrate cement – Test for physical properties – Different grades of cements – Admixtures – Mineral and chemical admixtures – accelerators, retarders, air entrainers, plasticizers, super plasticizers, fly ash and silica fume.

Aggregates: Classification of aggregate – Particle shape & texture – Bond, strength & other mechanical properties of aggregates – Specific gravity, Bulk density, porosity, adsorption & moisture content of aggregate – Bulking of sand –Deleterious substance in aggregate – Soundness of aggregate – Alkali aggregate reaction – Thermal properties – Sieve analysis – Fineness modulus – Gap graded and well graded aggregate as per relevant IS code – Maximum aggregate size.

UNIT – II

Fresh Concrete: Steps in Manufacture of Concrete–proportion, mixing, placing, compaction, finishing, curing – including various types in each stage. Properties of fresh concrete–Workability – Factors affecting workability – Measurement of workability by different tests, Setting times of concrete, Effect of time and temperature on workability – Segregation & bleeding – Mixing and vibration of concrete, Ready mixed concrete, Shotcrete.

UNIT – III

Hardened Concrete: Strength in tension & compression – Factors affecting strength – Relation between compression & tensile strength – Curing, Testing of Hardened Concrete: Compression tests – Tension tests – Factors affecting strength – Flexure tests – Splitting tests – Non-destructive testing methods – codal provisions for NDT.

UNIT – IV

Elasticity, Creep & Shrinkage– Modulus of elasticity – Dynamic modulus of elasticity – Poisson's ratio – Creep of concrete – Factors influencing creep – Relation between creep & time – Nature of creep – Effects of creep – Shrinkage –types of shrinkage.

UNIT – V

Mix Design: Factors in the choice of mix proportions– Quality Control of concrete – Statistical methods – Acceptance criteria – Concepts Proportioning of concrete mixes by various methods – ACI method of mix design, British DoE method of mix design and mix design as per IS 10262:2009.

Special Concretes: Ready mixed concrete, Shotcrete -Light weight aggregate concrete – Cellular concrete – No-fines concrete, High density concrete, Fibre reinforced concrete – Different types of fibres – Factors affecting properties of F.R.C, Polymer concrete – Types of Polymer concrete – Properties of polymer concrete, High performance concrete – Self consolidating concrete, SIFCON, self healing concrete.

Text Books:

1. Concrete Technology by M.S.Shetty. – S.Chand& Co.; 2017.
2. Concrete Technology by M.L. Gambhir. – Tata Mc. Graw Hill Publishers, New Delhi.

Reference Books:

1. Properties of Concrete by A.M.Neville – PEARSON – 4th edition.
2. Concrete Technology by A.R. Santha Kumar, Oxford University Press, New Delhi.

SURVEYING - II

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int.Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

Student shall be able to:

1. Learn and understand the different methods of Theodolite surveying.
2. Differentiate the Concepts of Tachometry Surveying.
3. Learn the different methods of Curve Setting.
4. Explore the methods of RS and GIS
5. Describe the procedures available in Total Station Instrument for surveying.

Course Outcomes:

After the completion of the course student should be able to

1. Appreciate the importance of Theodolite in Surveying
2. Apply Concepts of Tachometry in Surveying.
3. Construct the Curves in Highways, road construction and canal works.
4. Use the RS and GIS in designing
5. Use the Total Station in Surveying.

SYLLABUS**UNIT – I**

Theodolite- Types of Theodolite – Temporary Adjustments, Measurement of horizontal angle – Method of repetition, Method of reiteration – Uses of theodolites – Errors in theodolite or Permanent adjustments of a theodolite – Identification – Rectifying the errors.

UNIT - II

Traversing - Open and closed traverse – Closing errors, Balancing the error – Bowditch method – Transit method, Omitted measurements – Gales traverse table or Trigonometric levelling – Elevation of top of the tower - same plane - Different planes – Axis signal correction.

UNIT - III

Tacheometry– Principle of tacheometry – Stadia methods – Fixed hair method – Movable hair method – Tangential method – Subtense bar – Beam's stadia, Arc – Reduction diagrams or Triangulation – Classification-intervisibility of station – Signals and towers-baseline measurements – Corrections – Satellite station and Reduction to centre – Basenet.

UNIT – IV

Curves – Simple curves – Elements of simple curves – Methods of setting simple curves – Rankines method – Two theodolite method – Obstacles in curve setting – Compound curves– Elements of compound curves or Reverse curves – Elements of reverse curve –Determination of various elements – Transition curves – Ideal shape – Spiral transitioncurves - length of transition curve - Setting out methods.

UNIT- V

Total Station Surveying: Electronic Theodolite, Electronic Distance Measurements, TotalStation, Errors in measurements, Advantages, Disadvantages, Applications;

Modern surveying and mapping: GPS survey's – Introduction, Errors in GPS, Positioning methods, classification of GPS surveying, applications, advantages and disadvantages, photogrammetric surveying; sensors & platforms, aerial photogrammetry, orthophotography, topographic map, digital maps, DEM, GIS, Advantages & Disadvantages of photogrammetric surveying.

Text Books:

1. Dr. K.R. Arora, Surveying Vol. 1, 2, Standard Book House.
2. B.C.Punmia, Surveying Vol. 1, 2 and 3 Standard Book House.

Reference Books :

1. Surveying Vol. 1 and 2 – By S.K. Duggal, Tata Mc.Graw Hill Publishing Co.
2. Principles of GIS for land resource assessment by P.A. Burrough – Clarendon Press,Oxford.

REMOTE SENSING & GIS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The course is designed to

1. Introduce the basic principles of Remote Sensing and GIS techniques.
2. Learn various types of sensors and platforms
3. learn concepts of visual and digital image analyses
4. Understand the principles of spatial analysis
5. Appreciate application of RS and GIS to Civil engineering

Course Outcomes:

At the end of the course the student will be able to

1. Be familiar with ground, air and satellite based sensor platforms.
2. Interpret the aerial photographs and satellite imageries
3. Create and input spatial data for GIS application
4. Apply RS and GIS concepts in water resources engineering.

SYLLABUS**UNIT-I: Introduction to Remote Sensing Remote Sensing:**

Introduction, Basic components of remote sensing, electromagnetic radiation & electromagnetic spectrum and its interaction with atmosphere, energy interaction with the earth surfaces, Sensors: types and characteristics, passive sensor, active sensor, Platforms: airborne remote sensing & space borne remote sensing.

UNIT-II: Image Analysis

Introduction, elements of visual interpretations, Digital Image Processing - Image preprocessing, Image rectification, Image enhancement, Image classification: Supervised classification, Unsupervised classification.

UNIT-III: Geographic Information System(GIS)

Introduction, key components, application areas of GIS, Spatial data models: Raster data models, Vector data models, Raster versus Vector, Data input methods, Map projections.

UNIT-IV: RS & GIS Applications - General

Land Cover and Land Use, Agriculture, Forestry, Geology, Geomorphology, Urban applications.

UNIT-V: RS and GIS applications in Civil Engineering

Flood zoning and mapping, Groundwater prospects and Potential Recharge Zones, Watershed Management.Environmental Impact Assessment.

Text Books:

1. Remote and GIS by Basudeb Bhatta, Oxford Publications.
2. Remote Sensing and its Applications by LRA Narayana, Universities Press.
3. Basics of Remote Sensing and GIS by Dr. S. Kumar, Laxmi Publications (P) Ltd, New Delhi.
4. Remote Sensing and Geographical Information Systems by M. Anji Reddy, BS Publications.

Reference Books:

1. Concepts & Techniques of GIS by C.P.Lo Albert, K.W. Yong, Prentice Hall (India) Publications.
2. Principles of Geographical Information Systems by Peter A Burrough and Rachel A. Mc. Donnel, Oxford Publications
3. Fundamentals of Remote Sensing by George Joseph, University Press (India) Private Limited

SURVEYING FIELD WORK**Lab : 3 Periods****Exam : 3 Hrs.****Int. Marks : 50****Ext. Marks : 50****Credits : 2****Course Objectives:**

Student shall be able to

1. Understand and apply the basic methods of Chain Surveying and Compass Surveying.
2. Summarize the different methods of Plane Table Surveying
3. Observe and report the different types of Leveling
4. Study and identify the different methods of Theodolite surveying.
5. Describe the procedures available in Total Station Instrument for surveying.

Course Outcomes:

After the completion of the course student should be able to

1. Apply the linear measurement in simple Boundary Surveys.
2. Identify direction of any line using compass survey.
3. Relate the importance of Theodolite in Surveying
4. Apply Concepts of Tachometry in Surveying.
5. Use the Total Station in Surveying.

SYLLABUS

1. Chain Surveying: Finding the distance between inaccessible points by making use of chain, cross staff, tape, ranging rods; Arrows and field problems of obstacles to chaining.
2. Compass Survey: Finding the distance between inaccessible points by making use of compass, tape and ranging rods.
3. Plane Table Survey: Finding the distance between inaccessible points by making use of plane table, its accessories-Ranging rods and tape.
4. Levelling: Introduction to fly levelling-Booking the readings by height of collimation method and by rise and fall method-To find closing error.
5. Levelling: L.S. & C.S levelling for an open traverse
6. Theodolite: Distance between two in-accessible points by theodolite
7. Contour mapping using total station
8. Height of remote point using total station
9. Position of hidden point using total station
10. Area & volume measurement using total station

Reference Books:

1. B.C.Punmia , Ashok Kumar jain , Arun kr. Jain , Surveying I & II ,Laxmi publications

FLUID MECHANICS LAB - I

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course objectives:

1. To verify the principles of channel flow in laboratory by conducting experiments

Course Outcomes:

1. Define and Measure Fluid Properties.
2. Illustrate Flow Measuring Devices used in pipes, channels and Tanks
3. Analyze characteristics of broad crested weir.
4. Illustrate the characteristics of surface profiles in free and forced vibrations.
5. Compare sharp crested full width and contracted weirs

SYLLABUS

1. Study of Small orifice, by constant head method and Time of emptying a tank through a small orifice.
2. Study of Cylindrical mouthpiece by constant head method and Time of emptying a tank through a cylindrical mouthpiece.
3. Study of floating body and determination of Meta -centric Height.
4. Study of surface profiles in Free and Forced Vortex motions.
5. Study of Venturimeter.
6. Study of Orifice meter.
7. Study of Flow nozzle meter.
8. Study of Sharp – crested full width and contracted weirs.
9. Study of V-notch and Trapezoidal notch.
10. Study of Broad-crested weir.
11. Study of Frictional Resistance in pipes.
12. Study of types of flow in pipes

Reference Books:

1. Hydraulics Fluid Mechanics and Fluid Machines, S.Ramamrutham, Dhanpat Rai Publishing Co.
2. Engineering Fluid Mechanics by K.L. Kumar, S. Chand & Co
3. Fluid Mechanics and Hydraulic Machines by R.K. Bansal, Laxmi Publications.
4. F M White, Fluid Mechanics, Tata McGraw Hill Publication 2011.
5. Relevant NPTEL Courses.

INDUSTRY ORIENTED TECHNOLOGY LAB

Lab : 2 Periods
Exam : 3 Hrs.

Sessionals : 50
Credits : 1

**ADVANCED COMPUTATION SURVEYING / GEOINFORMATICS / GEOMATICS
ENGINEERING**

Course Objectives:

The course is design to fulfill the following objectives

1. To prepare the student to plan and conduct field work and application of scientific methodology in handling field samples.
2. To interpret the False Color Composite Images of Satellite data.
3. To Create Fundamental Knowledge on GIS software.
4. To learn the identification of Geographic Coordinates by GPS.
5. To Study the land use/ Land cover dynamics of certain region.

Course Outcomes:

Students who successfully complete this course will be able to:

1. Fully equipped with various surveying concepts and methods using advanced ground survey equipment's.
2. Carry out profiling and grid leveling, for generation of profiles, contour maps, and earth works computations.
3. Handle the Satellite images and interpret the satellite data.
4. The interpret data can be used to prepare plan for urban development/town planning.
5. Prepare the candidates with National Global employability.

SYLLABUS

1. Transferring and Drafting the collected raw data from total station survey using AutoCAD.
2. Computation on drafted data using AutoCAD.
3. Developing Contour using raw data from total station using surfer software.
4. Visual Interpretation of standard FCC (False color composite).
5. Digitization of physical features on a map / image using GIS software.
6. Coordinates measurement using GPS.
7. Field data collection under national land use / land cover mapping on 1:120000 scale using temporal AWIFS data.
8. Asset Mapping of village using Bhuvans Panchayat Moblie App.

Reference Books:

1. A M Chandra : Higher surveying
2. T M Lillesand et al: Remote sensing & Image Interpretation
3. B.Bhatta : Remote sensing&GIS
4. M.Anjireddy: Remote sensing & GIS,BS Publications
5. N K Agarwal : essentials of GPS,Spatial Networks, Hyderabad.

ENGLISH PROFICIENCY-II
(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

To equip the students with the components of the language required and help them gain adequate knowledge so as to become employable and competent.

Course Objectives:

Students will be able to

1. Enhance their Interpretative skills
2. Understand how to prepare a text.
3. Comprehend various types of writing discourses and respond.
4. Produce effectively different write-ups related to various business contexts.
5. Strengthen their emotional make-up.
6. Perceive various writing discourses.

Course Outcomes:

The students will

1. Develop the skills of taking and making notes
2. Interpret the pictures appropriately and effectively.
3. Read, comprehend and infer a given piece of writing effectively.
4. Learn and practice the skills of Research writing.
5. Communicate well through various forms of writing.
6. Be confident in giving presentations and dealing with people.

SYLLABUS

UNIT-1:SPEAKING

Analyzing proverbs

Enactment of One-act play

UNIT-2:READING

Reading Comprehension

Summarizing Newspaper Article

UNIT-3:WRITING

Note Taking &Note Making

Precis Writing

Essay Writing

Letter Writing

Picture Description

Literary Appreciation– Learning the Language of Literature

UNIT-4:VOCABULARY

Indian-origin English Words

Phrasal Verbs for Day-to-Day Communication

Commonly used Idiomatic Expressions

UNIT-5:PROJECT

Research Writing

Reference Books:

1. English for Colleges by Brendan J. Carroll, Macmillan Publications
2. Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw Hill
3. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
4. The Oxford Guide to Writing and Speaking by John Seely OUP
5. Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
6. English for Engineers and Scientists by Sangeeta Sharma and Binod Mishra, PHI Learning



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

CIVIL ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 CE 3101	Structural Analysis -II	3	3	1	--	4	30	70	100
B17 CE 3102	Reinforced Concrete Structures -I	3	3	1	--	4	30	70	100
B17 CE 3103	Steel Structures- I	3	3	1	--	4	30	70	100
B17 CE 3104	Geotechnical Engineering -I	3	3	1	--	4	30	70	100
B17 CE 3105	Fluid Mechanics -II	3	3	1	--	4	30	70	100
B17 CE 3106	Estimation & Quantity Surveying	3	3	1	--	4	30	70	100
B17 CE 3107	Environmental Engineering Lab	2	--	--	3	3	50	50	100
B17 CE 3108	Fluid Mechanics Lab II	2	--	--	3	3	50	50	100
B17 CE 3109	Geographic Information Systems Laboratory	1	--	--	2	2	50	50	100
B17BS 3101	Problem Solving & Linguistic Competence	1	--	3	--	3	30	70	100
B17BS 3105	IPR & Patents	--	--	2	--	2	--	--	--
Total		24	18	11	8	37	360	640	1000

STRUCTURAL ANALYSIS – II

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives: Students shall be able to understand:

1. Basic concept of Force method, and Displacement method and analysis of Statically indeterminate Trusses and statically indeterminate Rigid Frames.
2. Basic concept of Force method, and Displacement method and analysis of Statically indeterminate statically indeterminate Rigid Frames.
3. Analysis of Three hinged arch and Two hinged arch for different geometry and also explain Effects of rib-shortening and temperature change.
4. Analysis of cable subject different types of loads with Ends at same and different levels. Forces on anchor cable and tower, and analysis of suspension bridge with Two hinged and Three hinged Stiffening Girder.
5. To learn the concept of Matrix Methods of structural analysis.

Course Outcomes:

1. Student should be able to understand and analyse the statically indeterminate trusses by using method of consistent deformation and Castigliano's theorem – II
2. Analyse the statically indeterminate rigid frame by Slope Deflection Method, Moment Distribution Method, Kani's Methods and Column Analogy Method.
3. Analyze three hinged, two hinged arches and cables and suspension bridge with Two hinged and Three hinged Stiffening Girder.
4. Analyse suspension cables support are at the same level and different levels.
5. To develop stiffness matrix and flexibility matrix for two span continuous beams by direct method,

SYLLABUS**UNIT-I Analysis of statically indeterminate trusses**

Analysis of statically indeterminate trusses (having not more than 7 members and 3 supports) containing (a) external redundant supports (b) internal redundant members using (i) method of consistent deformation of unit load method (ii) Castigliano's theorem – II.

UNIT-II Analysis of statically indeterminate frames

Analysis of statically indeterminate Non Sway & Sway frames (single storey, single bay portal frames only) using (i) slope-deflection method (ii) moment distribution method (iii) Kani's method.

UNIT-III Arches

Arches: Introduction, Geometrical properties, Arch action, Normal thrust, radial shear and bending moment in three hinged and two hinged parabolic and segmental arches. Effects of yielding of supports, rib-shortening and temperature changes

UNIT-IV Cables and Suspension Bridges

Cables and Suspension Bridges: Introduction, Properties of a suspended cable, Stresses in loaded cables with supports at the same and different levels. Length of cable, support system, two & three hinged stiffening girders.

UNIT-V Matrix methods of structural analysis

Introduction to matrix methods of structural analysis (very elementary treatment only) static indeterminacy, kinematic indeterminacy, flexibility and stiffness method for two span and two degree continuous beams only.

Text Books:

1. Statically indeterminate structures – C.K. Wang , McGraw Hill Education PVT.LTD.
2. Structural analysis – A matrix approach – G.S. Pandit and S.P. Gupta.
3. Structural analysis – Devdas Menon, Narosa Publishing House PVT.LTD

Reference Books:

1. Basic structural Analysis- C S Reddy McGraw Hill Education PVT.LTD
2. Indeterminate Structures by R.L. Jindal
3. Mechanics of structures Vol. II- S.B. Junnarkar and Dr.H.J.shah, Charotar Publishing House.

REINFORCED CONCRETE STRUCTURES-I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives: Student shall be able to

1. Understand the different design philosophies on reinforced concrete and basic concept of Limit State of Collapse and learn how to analyze and design singly reinforced rectangular section, doubly reinforced rectangular section and flanged rectangular sections.
2. Understand concept behind the design of reinforced concrete beams subjected to flexure, shear and torsion and also design of bond.
3. Understand the behavior of one way and two way action in slabs and also design one way rectangular slab and two way rectangular slabs subjected to uniformly load.
4. Differentiate the short and long columns. Understand the concept behind in the design of short column subjected Axially loaded, Uniaxial moment and Biaxial moments.
5. Understand the concept behind the design of footing and staircase

Course Outcomes:

After the completion of the course student should be confident to

1. Analyze and design the flexural members.
2. Design the reinforced concrete beams subjected to shear only and also combined action of shear and torsion.
3. Distinguish between the behavior of one way and two way actions in slab and familiarize to design of two way slabs whose corners restrained and not restrained from lifting up.
4. Design the compression members.
5. Design the footing and staircase.

SYLLABUS

General: Loading standards as per IS 875, Grades of steel and cement, Introduction to basic design concepts like Working Stress Method (W.S.M), Ultimate Load Method (U.L.M) and Limit State Method (L.S.M.).

Limit State of Collapse: Introduction, Characteristic load and strengths, Design values, Partial safety factors Loads and materials, Stress-Strain characteristics of concrete and steel.

UNIT-I

Limit State of Collapse in Flexure: Assumption in analysis at Limit State of Collapse in Flexure, Limiting depth of neutral axis. Concrete stress block in compression. Under reinforced, Balanced and over reinforced sections. Analysis of singly reinforced rectangular sections, analysis of singly reinforced flanged section, analysis of doubly reinforced rectangular sections. Code requirement for design of flexural reinforcement are effective span, concrete cover, spacing of reinforcing bars, minimum and maximum areas of flexural reinforcement, requirements for deflection control, general guide lines for choosing beam size. Design of singly and doubly reinforced rectangular sections. Estimation of Effective flange width, Design of flanged beams (T-Beams),

UNIT-II

Limit State of Collapse in Shear: Shear stresses in homogeneous rectangular beam, modes of cracking, shear transfer mechanisms, shear span/depth ratio, shear failure modes. Calculation of nominal shear stress, critical sections for shear design, Design shear strength of concrete in beams with and without shear reinforcement. Types of shear reinforcement, Factors contributing to ultimate shear resistance, limiting ultimate shear resistance. Shear resistance of web reinforcement (Truss analogy), minimum shear reinforcement. Design of shear reinforcement in beams as per IS Code 456. Enhanced shear near supports. Steel detailing.

Limit State of Collapse in Torsion: Need for torsional reinforcement, Reinforcement for torsion in RC beams, design strength in torsion combined with flexure and IS code provisions for design of longitudinal reinforcement, design strength in torsion combined with shear and IS code provisions for design of transverse reinforcement, distribution of Torsional reinforcement. Design of rectangular section for combined bending shear and torsion. Detailing of torsion reinforcement

Limit State of Collapse in Bond: Concept of bond, Code requirement for bond, flexural bond, anchorage bond, development length. Bends, Hooks and Mechanical anchorages. Splicing of reinforcement, Lap Splices, Welded Splices and Mechanical Connections.

UNIT-III

Design of one way slabs : Behaviour of one-way slabs, general considerations for slabs, minimum flexural reinforcement in slabs, deflection control by limiting Span/Depth ratio. Design of simply supported one way slabs and design of continuous one way slabs using moment coefficient, detailing of reinforcement in one-way slabs.

Design of two way slab: Behaviour of two-way slabs, design of wall supported two-way slabs, slab thickness based on deflection control criterion, Uniformly loaded and simply supported rectangular slabs (Rankine-Grashoff theory), Uniformly loaded restrained rectangular slabs using IS code 456 provisions. Detailing of flexural reinforcement and torsional reinforcement. Shearforce in uniformly loaded two-way slabs.

UNIT-IV

Limit State of Collapse in Compression: Classification of columns based on type of reinforcement, type of loading and slenderness ratios. Estimation of effective length of a column- definition of effective length- unsupported length. Effective length ratios for idealized boundary conditions- Code recommendations for idealized boundary conditions. Code requirements on slenderness limits, minimum eccentricities and reinforcement. Design of short column under axial compression- condition of axial loading- behaviour under ultimate loads- Tied columns- Spiral columns. Design strength of axially loaded short columns. Design of Short columns subjected to combined axial load and uniaxial moments by using design hand book SP:16. Design of Short columns subjected to combined axial load and biaxial moments by using design hand book SP:16

UNIT-V

Design of Footings: Types of footing- Isolated footings, Soil pressures under isolated footings- Allowable soil pressure, Distribution of base pressure- Centrally loaded footings, Instability problems- Overturning and sliding. General design considerations and Code requirements- Factored soil pressure at ultimate limit state, general design considerations, Thickness of footing base slab, Design for Shear, design for Flexure, Transfer of force at Column Base. Design of isolated square and Rectangular footings concentrically loaded.

Design of Staircases: Types of staircases Geometrical Configurations, Structural Classification, stair slab spanning transversely, stair slab spanning longitudinally Loads and Load effects on stair Slabs, Design of stair slab spanning longitudinally.

Text Books:

1. Reinforced concrete design by S.Unnikrishna Pillai & Devdas Menon, Tata Mc.Graw Hill, New Delhi.
2. Limit State Design by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi, publications Pvt. Ltd., New Delhi .
3. Reinforced concrete Limit state design by Ashok K. Jain, Nem Chand & Bros, Roorkee
4. Limit state design of reinforced concrete – P.C.Varghese, Printice Hall of India, New Delhi.
5. Reinforced Concrete Vol.I (Elementary Reinforced concrete) by DR.H.J.Shah, Charotar publication house PVT.LTD, ANAND, Gujarath

Reference Books:

1. Fundamentals of Reinforced concrete design by M.L. Gambhir, Printice Hall of India Private Ltd., New Delhi.
2. Reinforced concrete structural elements – behaviour, Analysis and design by P.Purushotham, Tata Mc.Graw-Hill, 1994.
3. Design of concrete structures – Arthus H.Nilson, David Darwin, and Charles W. Dolar, Tata Mc.Graw-Hill, 3rd Edition, 2005.
4. Reinforced concrete structures, Vol.1, by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi, publications Pvt.Ltd., New Delhi
5. Reinforced concrete structures – I.C. Syal & A.K.Goel, S.Chand Publishers
6. Reinforced concrete design by N. Krishna Raju and R.N. Pranesh, New age International Publishers, New Delhi
7. Design of Reinforced Concrete Structures by N.Subramanian, Oxford University Press, YMCA Library Building, 1 Jail Singh Road, New Delhi, India.

STEEL STRUCTURES-I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Note: All the designs should be taught in the limit state design method as per IS 800-2007

Course Objectives:

Students shall be able to:

1. Understand the fundamental concepts of limit state design of steel structures; understand the bolted and welded connections, such as behavior of bolted joints.
2. Design strength of ordinary black bolts, high strength grip bolts.
3. Learn the simple connections, such as connections subjected to axial loads only.
4. Learn the welded connections, weld specifications, design of welded joints as per IS:800-2007 code.
5. Understand the behavior of steel structure under tension, compression and flexure and design the tension and compression members by using IS: 800-2007 code.

Course Outcomes: Student should be able to

1. Design of simple connections with bolted connections
2. Design of simple connections with welded connection,
3. Design of tension members subjected to axial force
4. Design of compression axially loaded members as built up columns
5. Design the laterally supported and unsupported beams as per I.S code.

SYLLABUS

UNIT-I

Fundamental Concepts of limit state design of structures, Different types of rolled steel sections available to be used in steel structures. Stress – Strain relationship for mild steel.

Bolted connections: Behaviour of bolted joints, Design strength of ordinary black bolts, high strength friction grip bolts, Simple connections.

UNIT-II

Welded Connections: Introduction ,welding processes, Advantages of welding, Types and properties of welds, Types of joints, weld specifications as per IS 800:2007 code provisions, Types of weld defects Design of lap joints and butt joints subjected to axial load by using fillet and butt welds.

UNIT-III

Tension members: Types of tension members, slenderness ratio, displacement of tension members, behaviour of tension members, modes of failure, factors affecting strength of tension members, design of tension members, Lug angles.

UNIT-IV

Compression members: Possible failure modes, behaviour of compression members, Effective length, radius of gyration and slenderness of compression members, Allowable stresses in compression, Design of axially loaded compression members, built up compression members with lateral supporting system such as lacing and battened.

UNIT-V

Beams: Beam types, section classifications, lateral stability of beams, Allowable stress in bending, Shear and Bearing stresses, Effective length of compression flange, Laterally supported and unsupported beams.

Text Book:

1. Design of Steel structures by N. Subramanian, Oxford University Press.
2. Limit State Design of steel structures – Ramchandra and VirendraGehlot, Scientific Publishers (India)
3. Limit State Design of steel structures by S.K.Duggal, McGraw Hill Education Private Ltd.

Reference Books:

1. Design of steel structures by K.S.Sai Ram, Pearson Education India.
2. Design of steel structures by Limit State Method as per IS: 800-2007 – S.S. Bhavikatti, IK International Publishing House, Bangalore – 560 001.

GEOTECHNICAL ENGINEERING-I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The objectives of this course are to make the student aware of

1. Fundamental relationships between different parameters of a soil mass.
2. Different types of soils and identify their properties.
3. Processes of compaction and consolidation and apply them to field problems.
4. Stress distribution, settlement and strength of different soils in different conditions.
5. Importance of Soil Mechanics in solving the engineering problems.

Course Outcomes:

Students will be able to

1. Know the fundamental relationships between different parameters of soil mass and classify different types of soils along with identifying their properties(K3)
2. Estimate Effective stresses and permeability of soils(K3)
3. Estimate stress distribution in soil for different Load conditions(K3)
4. Appreciate the processes of compaction and consolidation and apply them to field problems(K3)
5. Identify shear strength parameters for different conditions(K3)

SYLLABUS**UNIT-I**

Introduction: Historical development, Soil Formation, Minerals in clays and sand, Soil Structure, Physical properties of Soil: Void ratio, Porosity, Degree of Saturation, Water content, Specific Gravity, weight –volume Relationships, Relative density, Consistency limits: Determination and consistency indices, Activity, Sensitivity and Thixotropy. Mechanical analysis and Soil Classification: Sieve analysis, Stoke's law and hydrometer analysis. Unified soil classification, Indian Standard Soil Classification Systems, Field Identification of Soils

UNIT-II

Soil Hydraulics: Types of soil water, capillary rise and surface tension, Darcy's law and its limitations, constant head and variable head permeability tests, Factors effecting Coefficient of permeability, permeability of stratified soils. Total, neutral and effective stresses, Effective Stress Principle, Upward flow conditions, quick sand conditions and critical hydraulic gradient.

UNIT-III

Stress Distribution in Soils: Bousinesq's theory for determination of vertical stress, assumptions and validity, rectangular and circular loaded areas, Pressure Bulb and Influence diagrams, westergaard's theory, Newmarks influence chart - construction and use, 2:1 approximate method, contact pressure distribution beneath footings.

UNIT-IV

Compaction: Mechanism of compaction, Factors effecting compaction: water content, compaction effort, Type of soil. I.S Light and I.S Heavy compaction tests, Effect of compaction on soil Properties, Field compaction: compaction Equipment and Evaluation of field Compaction. **Consolidation:** Basic Definitions: compression index, coefficient of compressibility and coefficient of volume decrease. Terzaghi's one dimensional consolidation theory - assumption, Derivation of differential equation and Solution, Oedometer Test, Determination of coefficient of consolidation by time fitting methods, initial compression, primary compression and secondary compression, determination of preconsolidation pressure. Normally consolidated, over consolidated and under consolidated clays.

UNIT-V

Shear Strength of Soils: Stress at a point, Mohr circle of stress, Mohr coulomb failure theory, shear parameters, laboratory shear tests – shear box, triaxial and unconfined compression tests, laboratory and field vane shear tests, Sensitivity of clays, Types of shear tests on drainage conditions, shear strength of sands, critical void ratio and dilatancy, Factors affecting shear strength of clays and sands, Total stress analysis and Effective stress analysis

Text Books:

1. Soil Mechanics and Foundation Engineering by K.R. Arora.
2. Basic and Applied Soil Mechanics by GopalRajan and A.S.R. Rao.
3. Geotechnical Engineering by P. Purushothama Raj

Reference Books:

1. Principles of Geotechnical Engineering – BrajaM.Das

FLUID MECHANICS- II

Lecture : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

1. To understand the design philosophy of turbines and pumps.
2. To understand the fundamental concept for methods of dimensional analysis.
3. To know the design used for supplying water and generating power.

Course outcomes:

At the end of the course, the students will be able to:

1. Apply the principles of modeling pumps, turbines, propellers using various dimensionless numbers.
2. Determine discharge and design most economical channel section for uniform flow in open channels.
3. Use momentum and energy principles for design of turbines and pumps.
4. Recommend suitable type of turbines and pumps for the given project.

SYLLABUS**UNIT-I**

Dimensional Analysis and Similitude: Dimensional Homogeneity - Methods of Dimensional Analysis – Rayleigh’s Method – Buckingham’s π theorem – Superfluous and Omitted Variables - Similitude – Model Analysis – Dimensionless numbers – Similarity Laws – Model testing of partially submerged bodies – Types of models.

Boundary Layer Theory: Introduction – characteristics of laminar boundary layer – boundary layer growth over a flat plate (without pressure gradient) – Boundary thicknesses – Stability parameter – Turbulent boundary layer – boundary layer separation – Boundary layer on rough surfaces – laminar sub layer.

UNIT – II

Flow past submerged bodies: Introduction – Types of Drag – Drag on a sphere – Drag on a cylinder – Von Karman Vortex Trail – Drag on a flat plate – Development of Lift on immersed circular cylinder – Magnus effect.

Impact of Jets: Impulse momentum equation – Momentum Correction factor, Force on Stationary flat plate – moving flat plate - Force on Stationary curved vanes – moving curved vanes.

UNIT – III

Hydraulic Turbines: Introduction - Classification based on Head, Discharge, Hydraulic Action – Impulse and Reaction Turbines, Differences between Impulse and Reaction Turbine, Choice of Type of Turbine, Component Parts & Working principle of a Pelton Turbine, Francis Turbine - Velocity Triangles - Hydraulic and Overall efficiencies.

Performance of turbines: Performance under Unit head, power and speed – Performance under specific conditions - Specific Speed and its importance. Performance Characteristic Curves – Operating Characteristic Curves – Cavitation - Draft Tube.

UNIT – IV

Centrifugal Pumps: Types of Pumps – Selection Criterion – Comparison between Centrifugal & Reciprocating Pumps - Centrifugal Pumps – Component Parts & Working Principle – Classification of Centrifugal pumps - Cavitation – Maximum Suction lift – NPSH. Specific Speed of pumps – Performance Characteristics of Centrifugal Pumps – Dimensionless characteristics – Constant efficiency curves of Centrifugal Pumps

Reciprocating Pumps: Component Parts – Working Principle of single acting and double acting reciprocating pumps – Discharge Co-efficient, Volumetric efficiency and Slip. Work done and Power Input – Indicator Diagram, Effect of acceleration and friction on Indicator Diagram - Air Vessels.

UNIT – V

Flow through Open Channels: Classification of open channels, Uniform Flow: Chezy's and Manning's formula, Hydraulic mean depth, hydraulic radius. Most economical trapezoidal and rectangular channel section – Specific energy, Critical Flow.

Steady Rapidly Varied Flow: Hydraulic Jump in a horizontal rectangular channel, Specific force Computation of energy loss.

Text Books

1. Modi, P.N.& Seth, S.M. (2009), “Fluid Mechanics and Hydraulic Machinery”, Standard Book House, New Delhi, 19th Edition.
2. Jain, A.K. (2008), “Fluid Mechanics”, Khanna Publishers, New Delhi, 4th Edition.

Reference Books

1. Kumar, K.L., Chand, S. & Co. (2008), “Engineering Fluid Mechanics”, EurasiaPublishing House (P) Ltd, New Delhi, 8th Edition.
2. Subramanyam, K. (2008), “Flow in Open Channels”, McGraw Hill Education, New Delhi, 3rd Edition.
3. Chow, V.T. (2009), “Open-Channel Hydraulics”, The Blackburn Press, Caldwell, NJ USA, 1st Edition
4. White, F. M. (2011) “Fluid Mechanics”, Tata McGraw Hill Publication, New Delhi, 7th Edition
5. Relevant NPTEL Courses

ESTIMATION & QUANTITY SURVEYING

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

To provide the student with the ability to estimate the quantities of item of works involved in buildings, water supply, road works and irrigation works, and also to equip the student with the ability to do rate analysis, valuation of properties and preparation of reports for estimation of various items of work

Course Outcomes:

Students will be able to

1. List out various components, estimations and units of measurement for different works
2. Apply the method of building estimate to find out the quantities of various items of work
3. Determine the rate per unit of various items of work and their specifications
4. Explain the estimation of various roads and related items
5. Select various methods to find out the valuation of a property

SYLLABUS**UNIT-I**

Introduction: Definition, purpose and Importance of estimation, Standard units, Units of measurement of different items of work; Quantity surveying, accuracy and Errors in estimation; Different types of estimates, Data required for preparation of estimate, Different technical terms in estimation- Contingencies, Work charged Establishments, measurement book, schedule of rates and related terms in the estimate, different types of approvals. Plinth area and related terms used in the estimation of various structures, rules and methods of measurements of different works.

UNIT-II

Detailed estimate of buildings: Different items of work in building; Principles of taking out quantities, detailed measurement form; long wall and short wall method of building estimate, Centre line method of building estimate. Estimation of Load bearing wall and an RCC framed buildings; Bar bending schedule- Beams, Slabs

UNIT-III

Specifications : Meaning, purpose, types of specifications, general specification, detailed specifications of different items of buildings and other structures – Rate analysis –Data sheet for materials and various items of work in buildings and other structures, schedule of rates, abstract estimate of buildings.

UNIT-IV

Estimate of earth work in roads; different formulae for calculations, estimate of metalled road, Estimate of culverts

UNIT-V

Valuation of buildings; purpose, different methods of building valuation; different terms used in valuation and their meaning, valuation of government and private properties, Rent fixation

Text Books:

1. Estimating and Costing in Civil Engineering by B.N. Dutta.
2. Estimation, Costing, Specifications and Valuation in civil Engineering by M.Chakraborti.

Reference Books:

1. Textbook of estimating and costing by G.S. Birdie.
2. Textbook on Estimating, Costing and Accounts by D.D. Kohli and R.C. Kohli.

ENVIRONMENTAL ENGINEERING LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

The objectives of this course are to make the student aware of

1. Tests to identify the physical properties of water
2. Tests to identify the pH and conductivity of water
3. Tests to identify concentration of acidity and alkalinity in water
4. Tests to identify COD and BOD of water.

Course Outcomes:

Students will be able to:

1. Determine physical properties of water
2. Determine the turbidity and hardness of water
3. Determine COD and BOD of water
4. Estimate concentration of acidity and alkalinity
5. Estimate chloride content of water

SYLLABUS

List of experiments

1. pH
2. Electrical conductivity
3. Turbidity
4. Jar test
5. Hardness
6. Acidity
7. Alkalinity
8. Available chlorine and Residual chlorine
9. Estimation of total solids, suspended solids, dissolved solids
10. DO
11. BOD and COD
12. Chlorides

Reference Books:

1. Environmental Engineering by S.K.Garg

FLUID MECHANICS LAB– II**Lab : 3 Periods****Exam : 3 Hrs.****Int.Marks : 50****Ext. Marks : 50****Credits : 2**

Course Objectives:

The objectives of this course are to make the student aware of tests to identify the different types of flow measuring devices to measure flow in tanks, pipes and open channels.

Course Outcomes:

Students will be able to

1. Determine the coefficient of impact on a flat plate and curved vane by comparing the theoretical and actual forces by impact.
2. Analyze the working of the centrifugal pump and develop the characteristics of power input, head and efficiency under various discharges and plot the characteristic curves.
3. Analyze the working of the reciprocating pump and develop the characteristics of power input and discharge and efficiency under various heads and plot the characteristic curves.
4. Determine the performance characteristics of pelton wheel turbine and develop the characteristic curves of unit discharge, unit power and unit head under varying unit speed.
5. Determine the performance characteristics of Francis turbine and develop the characteristic curves of unit discharge, unit power and unit head under varying unit speed.

SYLLABUS

1. Study of Characteristics of a hydraulic jump – To measure and draw $(E_1 - E_2)/E_1$ vs F_1 and L_j/y_2 vs F_1 , and compare with theoretical results wherever possible.
2. Study of Rugosity coefficients in an open channel flow.
3. Study of Drag characteristics of a circular cylinder with its axis normal to the direction of flow.
 - (a) To measure the pressure distribution on the surface of a cylinder and plot the dimensionless pressure variation around the cylinder and compute the pressure drag.
 - (b) To measure the velocity variation in the wake of the cylinder, velocity of approach, and compute the total drag by momentum principle.
4. Study of performance characteristics of a centrifugal pump – To measure the discharge, head developed, and power input at various discharges for centrifugal pump and draw the performance characteristics.
5. Study of performance characteristics of a reciprocating pump – To measure the discharge, head developed, and power input at various discharges for reciprocating pump and calculate percentage slip and efficiency.
6. Study of performance characteristics of a Pelton turbine – To measure the discharge, head difference across the turbine, the brake load, speed of turbine for various discharges and draw the performance characteristics.
7. Study of performance characteristics of a Francis turbine – To measure the discharge, head difference across the turbine, the brake load, speed of turbine for various discharges and draw the performance characteristics.
8. Study of impact of a jet on flat and curved vanes.

Reference Books:

1. Fluid Mechanics with Laboratory Manual by B. Majumdar

GEOGRAPHIC INFORMATION SYSTEMS LABORATORY

Lab : 2 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 1

Course Objectives:

The course is designed to

1. Familiarize the students with digital map creation and updation life cycle
2. Introduce and make students practice various satellite image processing techniques
3. Make students practice the process of digitization, creation of thematic map
4. Introduce the students to advanced tasks like 3D modelling using spatial data.

Course outcomes:

At the end of the course, the student will be able to

1. Assign appropriate datum and projection systems for the given data
2. Preprocess the raw data to make it suitable for overlaying with various themes
3. Create thematic layers by using digitization techniques and attaching attribute data
4. Visualize and interpret digital elevation model

EXCERCISES IN GIS:

1. Digitization of Map/Toposheet
2. Creation of thematic maps.
3. Estimation of features and interpretation
4. Developing Digital Elevation model

GIS:

SOFTWARES:

1. ArcGIS 10.1
2. ERDASImagine9.3

Reference Books:

1. Principles of Geographical Information Systems by Peter A Burrough and Rachel A. Mc. Donnel, Oxford Publications
2. Concepts & Techniques of GIS by C.P.Lo Albert, K.W. Yong, Prentice Hall (India) Publications

PROBLEM SOLVING & LINGUISTIC COMPETENCE**(Common to all Branches)**

Tutorial	: 3 Periods (VA-2+QA-1)	Int.Marks	: 30
Exam	: 3 Hrs.	Ext.Marks	: 70
		Credits	: 1

Part-A: Verbal and Soft Skills-I**Course Objectives:**

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS**Grammar: (VA)**

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms.

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cambridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

IPR & PATENTS
(Common to CE, EEE & ME)

Tutorial : 2 Periods

Credits : 0

Course Objectives:

1. To introduce the idea of tangible and intangible property and its protection.
2. To familiarize with the frameworks for protection of intellectual property.
3. To layout the procedures to claim rights over intellectual property.

Course Outcomes:

After successful completion of the course, the student shall be able to

1. Identify various types of intangible property that an engineering professional could generate in the course of his career.
2. Distinguish between various types of protection granted to Intellectual Property such as Patents, Copy Rights, Trademarks etc.,
3. List the steps involved in getting protection over various types of intellectual property and maintaining them.
4. Take precautions in writing scientific and technical reports without plagiarism.
5. Help micro, small and medium entrepreneurs in protecting their IP and respecting others IP as part of their business processes.

SYLLABUS

UNIT I

Intellectual Property Law: Basics - Types of Intellectual Property - Innovations and Inventions - Trade related Intellectual Property Rights – Agencies Responsible for Intellectual Property Registration – Infringement - Compliance and Liability Issues

UNIT II

Principles of Copyright – Subject Matters of Copyright – Rights Afforded by Copyright Law – Copyright Ownership–Copyright Formalities and Registration – Limitations – Infringement of Copyright - Plagiarism and difference between Copyright infringement and Plagiarism

UNIT III

Introduction to Trade Mark – Trade Mark Registration Process – Post registration procedures – Trade Mark maintenance– Infringement – Dilution of Ownership of Trade Mark – Likelihood of confusion – Trade Mark claims – Trade Marks Litigation – International Trade Mark Law

UNIT IV

Introduction to Patent Law – Rights and Limitations – Rights under Patent Law – Patent Requirements – Ownership and Transfer – Patent Application Process and Granting of Patent – Patent Infringement and Litigation – International Patent Law – Double Patenting

UNIT V

Introduction to Trade Secrets – Maintaining Trade Secret – Physical Security – Employee Access Limitation – Employee Confidentiality Agreement – Trade Secret Law – Unfair Competition – Trade Secret Litigation – Breach of Contract – Applying State Law.

Text Books:

1. Kompal Bansal & Parikshit Bansal "Fundamentals of Intellectual Property for Engineers", BS Publications
2. Prabhuddha Ganguli: "Intellectual Property Rights" Tata McGraw –Hill, New Delhi
3. R. Radha Krishnan, S. Balasubramanian: "Intellectual Property Rights: Text and Cases", Excel Books, New Delhi.

Reference Books:

1. Deborah E. Bouchoux: "Intellectual Property". Cengage learning, New Delhi
2. Richard Stim: "Intellectual Property", Cengage Learning, New Delhi.

SCHEME OF INSTRUCTION & EXAMINATION
 (Regulation R17)
III/IV B.TECH
 (With effect from **2017-2018** Admitted Batch onwards)
CIVIL ENGINEERING
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 CE 3201	Reinforced Concrete Structures-II	3	3	1	--	4	30	70	100
B17 CE 3202	Steel Structures-II	3	3	1	--	4	30	70	100
B17 CE 3203	Geotechnical Engineering-II	3	3	1	--	4	30	70	100
B17 CE 3204	Transportation Engineering-I	3	3	1	--	4	30	70	100
B17 CE 3205	Air pollution and Control	3	3	1	--	4	30	70	100
#OE	Open Elective	3	3	1	--	4	30	70	100
B17 CE 3209	Geotechnical Engineering Lab	2	--	--	3	3	50	50	100
B17 CE 3210	Concrete Technology Lab	2	--	--	3	3	50	50	100
B17BS 3201	Employability Skills	1	--	3	--	3	30	70	100
B17BS 3202	Basic Coding	1	--	--	3	3	50	50	100
Total		24	18	9	9	36	360	640	1000

Open Elective	B17CS3213	Data Base Management Systems
	B17CE3206	Alternative Energy Sources
	B17CE3207	Wastewater Management
	B17CE3208	Green Fuel Technologies

REINFORCED CONCRETE STRUCTURES – II

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives: Student shall be able to

1. Understand the design concepts of cantilever and counterfort retaining walls.
2. Understand the design concept of reinforced concrete water tanks.
3. Understand the design concept of reinforced concrete bridges.
4. Understand behavior of stair slab spanning transversely and longitudinally and also design concept of piles and pile cap
5. Understand the design concept of flat slabs.

Course Outcomes:

After the completion of the course student should be confident to

1. Distinguish between the behavior of cantilever and counterfort retaining walls and design the cantilever and counterfort retaining walls.
2. Design the reinforced concrete circular and rectangular water tanks.
3. Design the reinforced concrete T-beam bridge.
4. Design the piles and pile cap.
5. Design the flat slabs

SYLLABUS**UNIT I**

Design of Retaining Walls: Types of retaining walls and their behavior, forces on retaining walls, Theories of Earth pressures-Rankine's and Coulomb's earth pressure theories (c and ϕ soils). Earth pressures and Stability requirements-Lateral earth pressures, Effect of surcharge on a level Backfill, Effect of water in the backfill. Stability requirements. Soil bearing pressure requirements. Drainage of retaining walls. Proportioning and Design of cantilever and counterfort retaining walls- position of Stem on base slab for economical design. Proportioning and Design of Elements of cantilever wall- Thickness of Base slab and Stem, design of stem, Toe and Heel Slab. Proportioning and Design of Elements of a counterfort wall- Thickness of various elements, design of stem, Toe and Heel Slab, Design of counterforts.

UNIT II

Water Tanks: Classification-Basis of Design-Permissible stress in concrete and steel in water tanks-Joints in tanks- Flexible joint-Joints of bottom slabs of tanks-Joints between wall and floor-Tanks resting on ground-Circular tank on ground with flexible joint between wall and the base slab Approximate design of circular tanks with walls restrained at the base-Rectangular tanks on ground-Analysis of a tank wall section subjected to bending moment and pull-Overhead tanks- Rectangular tank supported on masonry walls all round -Wind load analysis of columns and bracings of water tower-design of a square tank supported on columns and beams - Underground rectangular tanks.

UNIT III

Bridges: Components of a bridge in sub structure and super structure. Classification of bridges. Highway loading standards, kerbs, footpaths, railings, parapet loadings, Impact, wind, longitudinal forces. Design of solid slabs (casual reference to MOST drawings) Design of T-beam bridge deck slab, Longitudinal and Cross beams (casual reference to MOST drawings) Courbon's theory.

UNIT IV

Design of Piles: Behaviour of piles. Static formula for pile capacity, dynamic pile formula, Pile groups. Structural design of piles-design of bored cast in situ piles (bearing and friction types), under reamed piles.

Design of Pile caps: Code requirements for pile cap design Sectional method of design of pile cap, Strut-and-tie model for pile caps, Detailing of pile caps.

UNIT V

Design of Flat Slabs: Introduction, Proportioning of Flat slabs- Thickness of flat slabs, Drop panel, Column Heads, Shear Caps, Behaviour of flat slabs, Methods of Analysis-Direct Design Method, Equivalent Frame method, Transfer of Moments to column. Shear in Flat slabs- One-way and Two-way shear. Design procedure for flat slabs, Detailing of reinforcement.

Text Books:

1. Reinforced concrete design by S.Unnikrishna Pillai & Devdas Menon, Tata Mc.Graw Hill, New Delhi.
2. Limit State Design by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi, publications Pvt. Ltd., New Delhi.
3. Reinforced concrete Limit state design by Ashok K. Jain, Nem Chand & Bros, Roorkee.
4. Limit state design of reinforced concrete – P.C.Varghese, Printice Hall of India, New Delhi.
5. Reinforced Concrete Vol.I (Elementary Reinforced concrete) by DR.H.J.Shah, Charotar publication house PVT.LTD, ANAND, Gujarath.

Reference Books:

1. Fundamentals of Reinforced concrete design by M.L.Gambhir, Printice Hall of India Private Ltd., New Delhi.
2. Reinforced concrete structural elements-behaviour, Analysis and design by P.Purushotham, Tata Mc.Graw-Hill, 1994.
3. Design of concrete structures – Arthus H.Nilson, David Darwin, and Charles W. Dolar, Tata Mc.Graw-Hill, 3rd Edition, 2005.
4. Reinforced concrete structures, Vol.1, by B.C.Punmia, Ashok Kumar Jain and Arun Kumar Jain, Laxmi, publications Pvt.Ltd., New Delhi.
5. Reinforced concrete structures – I.C. Syal & A.K.Goel, S.Chand Publishers
6. Design of Reinforced Concrete Structures by N.Subramanian, Oxford University Press, YMCA Library Building, 1 Jail Singh Road, New Delhi, India.

STEEL STRUCTURES-II

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives: Students shall be able to:

1. Understand the design of columns bases.
2. Understand various components of a plate girder, design components of a plate girder by using IS code.
3. Understand use of IS code provisions for designing the vertical stiffeners, bearing stiffeners etc.
4. Understand design of beam-column connections subjected to eccentric shear connections.
5. Understand the designing of elevated circular steel tanks in working stress method.

Course Outcomes: Student should be able to

1. Design of columns bases.
2. Design components of a plate girder with and without stiffeners by using IS: 800-2007 code
3. Design of circular water tank in working stress method.
4. Design of beam-column connections subjected to eccentric shear connections.
5. Design of end bearings

SYLLABUS**UNIT-I**

Column bases: Introduction, Types of column bases, Theoretical considerations, Allowable stress in bearing, Design of Slab base, Design of Gusset base as per IS 800: 2007.

UNIT-II

Plate Girders (Welded): Components of a plate girder, Economical depth, proportioning of web and flanges, shear buckling resistance of web by simple post critical and tension field methods, connection of flange angles to web and flange angles to flange plates. Web stiffeners: Intermediate Transverse Stiffeners and longitudinal stiffeners, Design of bearing stiffeners. End panel design, design of intermediate Transverse stiffeners, connections, Curtailment of Flanges.

UNIT-III

Water tanks: Introduction, permissible stresses in water tanks, Design of circular steel water tank: forces acting over the tanks, stresses in elevated circular tanks, stresses in segmental and spherical bottoms, stresses in conical bottom, Design of circular girder (ring beam), staggering for circular steel tanks, stresses in columns, wind bracings.

UNIT-IV

Eccentric shear connections: Introduction, beam-columns connections, connections subjected to eccentric shear: welded seat connections: unstiffened seat angle connection and stiffened seat angle connections.

UNIT V

Bearings: Introduction, IS Code requirement for bearings, types of bearings, permissible stresses on bearings, Design of rocker and roller bearings using working stress method.

Text Books:

1. Design of Steel structures by N. Subramanian, Oxford University Press.
2. Limit State Design of steel structures – Ramchandra and Virendra Gehlot, Scientific Publishers (India)
3. Limit State Design of steel structures by S.K. Duggal, McGraw Hill Education Private Ltd.

Reference Books:

1. Design of steel structures by K.S. Sai Ram, Pearson Education India.
2. Design of steel structures by Limit State Method as per IS: 800-2007 – S.S. Bhavikatti, IK International Publishing House, Bangalore – 560 001.

GEOTECHNICAL ENGINEERING-II

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The objectives of this course are to make the student aware of

1. Different techniques of soil exploration.
2. Various methods for estimating bearing capacity of different types of foundations.
3. Estimating Load capacity of single piles and groups of piles.
4. Settlement analysis of foundations.
5. Theoretical aspects of well foundations.
6. Concepts of stability of finite and infinite slopes.
7. Earth pressures on retaining walls by Rankine's and Coulomb's theories.

Course Outcomes:

Students will be able to

1. Plan a detailed soil exploration programme (K2).
2. Apply various methods for estimating bearing capacity of different types of foundations. (K3)
3. Estimate load capacity of single piles and groups of piles and know the theory aspects of well foundations (K3)
4. Determine the stability of finite and infinite slopes. (K3)
5. Calculate earth pressures on retaining walls using Rankine's and Coulomb's theories (K3)

SYLLABUS**UNIT-I**

Subsoil Exploration: Methods of subsoil exploration Direct, semi direct and indirect methods, Soundings by Standard, Dynamic cone and static cone penetration tests, Types of Boring, Types of samples, Criteria for undisturbed samples, Transport and preservation of samples, Bore logs, planning of exploration programmes, report writing.

UNIT-II

Shallow Foundations: Factors effecting location of foundation and design considerations of shallow foundations. **Bearing Capacity:** Safe bearing capacity and allowable bearing pressure, General and local shear failures, Terzaghi's bearing capacity equations its modifications for square, rectangular and circular foundations, Factors affecting bearing capacity of Soil, Effect of water table on bearing capacity, IS Code method for Bearing capacity of footings, Allowable bearing pressure based on N-values, Field plate load tests, Settlement Analysis of shallow foundations

UNIT-III

Pile Foundations: Types, Construction, load carrying capacity of single piles in sands and clays (α -method) Dynamic Formula, Static formula, Pile load tests, Load carrying capacity of pile groups, Settlement Analysis of pile foundations. **Caissons:** Types of caissons, pneumatic caissons, Different shapes of well foundations, Relative advantages and disadvantages, Different Components of well and their function, Grip length, problems in well sinking and remedial measures

UNIT-IV

Stability Analysis of Slopes: Infinite and Finite Slopes, Stability Analysis of Infinite Slopes, different factors of safety, Types of Slope Failures – Toe, slope and Base failure, Stability Analysis of Finite slopes – Swedish Circle method, Friction Circle method, Fellenius method for location of Critical Slip Circle, Taylor's stability number

UNIT-V

Earth Pressure: Types of Earth pressure, Rankine's Active and passive earth pressure, Smooth Vertical wall with horizontal and inclined backfills. Coloumb's wedge theory, Culmans active earth pressure.

Text Books:

1. Basic and Applied Soil Mechanics by GopalRanjan and A.S.R. Rao
2. Foundation Analysis and Design – J. E. Bowles.
3. Soil Mechanics and Foundation Engineering – By K.R. Arora

Reference Books:

1. Principles of Foundation Engineering- BrajaM.Das
2. Foundation Engineering- P.C.Verghese

TRANSPORTATION ENGINEERING - I

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives: The objectives of this course are:

1. To impart different concepts in the field of Highway Engineering.
2. To acquire design principles of Highway Geometrics and Pavements
3. To learn various highway construction and maintenance procedures

Course Outcomes: Upon the successful completion of this course, the students will be able to:

1. Plan highway network for a given area.
2. Determine Highway alignment and design highway geometrics
3. Design Intersections and prepare traffic management plans
4. Judge suitability of pavement materials and design flexible and rigid pavements
5. Construct and maintain highways

SYLLABUS**UNIT-I**

Highway Planning and Alignment: Highway development in India; Classification of Roads; Road Network Patterns; Necessity for Highway Planning; Different Road Development Plans

– First, second, third road development plans, road development vision 2021, Rural Road Development Plan – Vision 2025; Planning Surveys; Highway Alignment- Factors affecting Alignment- Engineering Surveys – Drawings and Reports.

UNIT-II

Highway Geometric Design: Importance of Geometric Design- Design controls and Criteria- Highway Cross Section Elements- Sight Distance Elements- Stopping sight Distance, Overtaking Sight Distance and Intermediate Sight Distance- Design of Horizontal Alignment- Design of Super elevation and Extra widening- Design of Transition Curves- Design of Vertical alignment- Gradients- Vertical curves.

UNIT-III

Traffic Engineering: Basic Parameters of Traffic- Volume, Speed and Density- Traffic Volume Studies; Speed studies – spot speed and speed & delay studies; Parking Studies; Road Accidents- Causes and Preventive measures - Condition Diagram and Collision Diagrams; PCU Factors, Capacity of Highways – Factors Affecting; LOS Concepts; Road Traffic Signs; Road markings; Types of Intersections; At-Grade Intersections – Design of Plain, Flared, Rotary and Channelized Intersections; Design of Traffic Signals – Webster Method – IRC Method.

UNIT-IV

Highway Materials: Subgrade soil: classification –Group Index – Subgrade soil strength – California Bearing Ratio – Modulus of Subgrade Reaction. Stone aggregates: Desirable properties – Tests for Road Aggregates – Bituminous Materials: Types – Desirable properties – Tests on Bitumen – Bituminous paving mixes: Requirements – Marshall Method of Mix Design.

Design Of Pavements:Types of pavements; Functions and requirements of different components of pavements; Design Factors

Flexible Pavements:Design factors – Flexible Pavement Design Methods – CBR method – IRC method – Burmister method – Mechanistic method – IRC Method for Low volume Flexible pavements.

Rigid Pavements:Design Considerations – wheel load stresses – Temperature stresses – Frictional stresses – Combination of stresses – Design of slabs – Design of Joints – IRC method – Rigid pavements for low volume roads – Continuously Reinforced Cement Concrete Pavements – Roller Compacted Concrete Pavements.

UNIT-V

Highway Construction and Maintenance: Types of Highway Construction – Earthwork; Construction of Earth Roads, Gravel Roads, Water Bound Macadam Roads, Bituminous Pavements and Construction of Cement Concrete Pavements.

Pavement Failures, Maintenance of Highways, pavement evaluation, strengthening of existing pavements

Text Books:

1. Highway Engineering, Khanna S. K., Justo C. E. G and Veeraragavan A, Nem Chand Bros.,Roorkee.
2. Traffic Engineering and Transportation Planning, Kadiyali L. R, Khanna Publishers, NewDelhi.

Reference Books:

1. Principles of Highway Engineering, Kadiyali L. R, Khanna Publishers, NewDelhi
2. Principles of Transportation Engineering, ParthaChakroborthy and Animesh Das, PHI Learning Private Limited, Delhi
3. Highway Engineering, Paul H. Wright and Karen K Dixon, Wiley Student Edition, Wiley India (P) Ltd., New Delhi
4. Transportation Engineering - An Introduction, JotinKhisty C, Prentice Hall, Englewood Cliffs, New Jersey.
5. Traffic & Highway Engineering by Nicholas J. Garber, Lester A. Hoel, Fifth Edition, published in 2015,CENGAGE Learning, New Delhi.
6. Transportation Engineering and Planning, Papacostas C.S. and P.D. Prevedouros, Prentice Hall of India Pvt.Ltd; New Delhi.
7. Highway Engineering, Srinivasa Kumar R, Universities Press, Hyderabad
8. Practice and Design of Highway Engineering, Sharma S. K., Principles, S. Chand & Company Private Limited, New Delhi.
9. Highway and Traffic Engineering, Subhash C. Saxena, CBS Publishers, New Delhi.
10. Transportation Engineering Volume I by C. Venkatramaiah, Universities Press, New Delhi.

AIR POLLUTION AND CONTROL

Lecture : 3 Periods
Tutorial : 1 Period
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

Student shall be able to:

1. Explain the sources, characteristics, effects of air pollution
 2. Understand the meteorological conditions, plume behaviour
- Understand the various pollution control equipment's and their processes

Course Outcomes:

By the end of the course, the student should be able to:

1. Explain the classification, sources and effects of air pollution [K2]
2. Explain the different meteorological conditions that influence the dispersion of the pollutants [K2]
3. Illustrate the plume behaviour for different atmospheric stability conditions [K2]
4. Adapt various pollution control equipment's or methods to control the discharge of pollutants [K3]
5. Measure the pollution levels by sampling and analysis [K5].

SYLLABUS**UNIT-I**

Air Pollution and its definition – Factors influencing air pollution – Classification of pollutants particulates – Gases – sources of pollution – Air qualities standards – effects – Location of industries.

UNIT-II

Meteorology – Wind roses – lapse rates – mixing depth atmospheric dispersion – plume behaviour accumulation, estimation of pollutants – effective stack height.

UNIT-III

Air pollution effects on human beings, animals, plants and materials – Air pollution Episodes in India and Abroad.

UNIT-IV

Air Pollution Sampling and measurement Ambient air quality monitoring and stack monitoring, collection of particulate and gaseous pollutants: Isokenitic sampling: Ambient air quality survey.

UNIT-V

Control of air pollution – removal of pollutants – particulate and gaseous – Air pollution control equipments (units) such as settling chamber, cyclones, wet scrubbers/collectors, scrubbers, centrifugal scrubbers, spray towers, packed beds, electrostatic precipitators, after burners – absorption – adsorption – diffusion.

Text Books:

1. Air pollution Control Technology by C.S.Rao.
2. Air Pollution Control by K.V.S.G. Murali Krishna

Reference Books:

1. Elements of Air Pollution Control by Prof. T. ShivajiRao.
2. Fundamentals of Air Pollution by Dr. B.S.N. Raju, Oxford & I.B.H.

DATABASE MANAGEMENT SYSTEMS
(Common to CE & ME)
(Open elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand the basic concepts and the applications of database systems.
2. To master the basics of SQL and construct queries using SQL.
3. To understand the relational database design principles.
4. To become familiar with the basic issues of transaction processing and concurrency control.
5. To become familiar with database storage structures and access techniques.

Course Outcomes:

1. Demonstrate the basic elements of a relational database management system.
2. Ability to identify the data models for relevant problems.
3. Ability to design entity relationship model and convert entity relationship diagrams into RDBMS and formulate SQL queries on the data.
4. Apply normalization for the development of application software

SYLLABUS**UNIT-I**

Introduction: Database System Applications, Purpose of Database Systems, View of Data, Database Languages – DDL, DML, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Data Mining and Information Retrieval, Specialty Databases, Database Users and Administrators, History of Database Systems. Introduction to Data base design: Database Design and ER diagrams, Entities, Attributes and Entity sets, Relationships and Relationship sets, Additional features of ER Model, Conceptual Design with the ER Model, Conceptual Design for Large enterprises. Relational Model: Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity constraints, Querying relational data, Logical data base Design: ER to Relational, Introduction to Views, Destroying /Altering Tables and Views.

UNIT-II

Relational Algebra and Calculus: Preliminaries, Relational Algebra, Relational calculus – Tuple relational Calculus, Domain relational calculus, Expressive Power of Algebra and calculus. SQL: Queries, Constraints, Triggers: Form of Basic SQL Query, UNION, INTERSECT, and EXCEPT, Nested Queries, Aggregate Operators, NULL values Complex Integrity Constraints in SQL, Triggers and Active Data bases, Designing Active Databases..

UNIT-III

Schema Refinement and Normal Forms: Introduction to Schema Refinement, Functional Dependencies-Reasoning about FDs, Normal Forms, Properties of Decompositions, Normalization, Schema Refinement in Database Design, Other Kinds of Dependencies.

UNIT-IV

Transaction Management: Transactions, Transaction Concept, A Simple Transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Serializability, Transaction Isolation and Atomicity Transaction Isolation Levels, Implementation of Isolation Levels. Concurrency Control: Lock-Based Protocols, Multiple Granularity, Timestamp-Based Protocols, Validation-Based Protocols, Multiversion Schemes. Recovery System-Failure Classification, Storage, Recovery and Atomicity, Recovery Algorithm, Buffer Management, Failure with loss of nonvolatile storage, Early Lock Release and Logical Undo Operations, Remote Backup systems.

UNIT-V

Storage and Indexing: Overview of Storage and Indexing: Data on External Storage, File Organization and Indexing, Index Data Structures, Comparison of File Organizations. Tree-Structured Indexing: Intuition for tree Indexes, Indexed Sequential Access Method (ISAM), B+ Trees: A Dynamic Index Structure, Search, Insert, Delete. Hash- Based Indexing: Static Hashing, Extendible hashing, Linear Hashing, Extendible vs. Linear Hashing.

Text Books:

1. Data base Management Systems, Raghu Ramakrishnan, Johannes Gehrke, McGraw Hill Education (India) Private Limited, 3rd Edition.
2. Data base System Concepts, A. Silberschatz, Henry. F. Korth, S. Sudarshan, McGraw Hill Education(India) Private Limited l, 6th edition

Reference Books:

1. Database Systems, 6th edition, R Elmasri, ShamkantB.Navathe, Pearson Education.
2. Database System Concepts, Peter Rob & Carlos Coronel, Cengage Learning.
3. Introduction to Database Management, M. L. Gillenson and others, Wiley Student Edition.
4. Database Development and Management, Lee Chao, Auerbach publications, Taylor & Francis Group. Introduction to Database Systems, C. J. Date, Pearson Education.

ALTERNATIVE ENERGY SOURCES (Open elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives:

Student shall be able to:

1. Explain the concepts of Non-renewable and renewable energy systems
2. Outline utilization of renewable energy sources for both domestic and industrial applications
3. Analyze the environmental and cost economics of renewable energy sources in comparison with fossil fuels.

Course outcomes:

By the end of the course, the student should be able to:

1. Summarize the need of renewable sources in Global scenario [K2]
2. Explain the solar thermal conversion processes [K2]
3. Explain the wind energy conversion techniques [K2]
4. Explain the biomass energy conversion methodologies [K5]
5. Analyze the principle of ocean thermal energy conversion system [K4]

SYLLABUS

UNIT – I

Global and National Energy Scenario: Over view of conventional & renewable energy sources - need & development of renewable energy sources - Future of Energy Use, Energy for sustainable development - Potential of renewable energy sources - renewable electricity and key elements - Global climate change - CO₂ reduction potential of renewable energy - concept of Hybrid systems.

UNIT – II

Solar Energy: Solar energy system - Solar Radiation – Availability - Measurement and Estimation - Solar Thermal Conversion Devices and Storage - Applications Solar Photovoltaic Conversion, applications of solar energy systems.

UNIT – III

Wind Energy: Wind Energy Conversion - Site selection, Types of wind turbines, wind Generation and Control. Nature of the wind, , factors influencing wind, wind data and energy estimation, wind speed monitoring, classification of wind, characteristics, applications of wind turbines, offshore wind energy – Hybrid systems, wind resource assessment, Betz limit, site selection, wind energy conversion devices. Wind mill component design, economics and demand side management, energy wheeling, and energy banking concepts. Safety and environmental aspects, wind energy potential and installation in India.

UNIT – IV

Biogas: Calorific value and composition of biogas – Bio energy systems – Biomass conversion processes – Thermo chemical conversion processes – biomass gasification – pyrolysis – liquefaction – anaerobic digestion – Urban waste to energy conversion – bio diesel production – Biomass energy programme in India.

UNIT – V

Ocean Energy – Principle of Ocean Thermal Energy Conversion (OTEC) – tidal energy conversion – Scheme of development of tidal energy

Hydro power plants- types of turbines – estimation of primary and secondary power

Geothermal Energy – Geothermal power plants

Text Books:

1. Non-Conventional Energy Sources by G.D.Rai
2. Twidell, J.W. and Weir, A., Renewable Energy Sources, EFN Spon Ltd., 1986.

Reference books:

1. Kishore VVN, Renewable Energy Engineering and Technology, Teri Press, New Delhi, 2012
2. Godfrey Boyle, Renewable Energy, Power for a Sustainable Future, Oxford University Press, U.K., 1996.

WASTEWATER MANAGEMENT
(Common to CE & ME)
(Open elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives:

Student shall be able to:

1. Explain the difference between domestic waste and industrial waste water
2. Understand the common methods of treatment used in industries
3. To understand the effluent quality and its impact on environment.

Course outcomes:

By the end of the course, the student should be able to:

1. Define the quality of industrial wastes [K1]
2. Explain various industrial waste treatment processes [K2]
3. Outline the advanced treatment techniques available for industrial wastes [K2]
4. Explain the sludge reduction and disposal methodologies [K2]
5. Analyse the waste effluent treatment from different case studies [K4].

SYLLABUS

UNIT – I

Water uses by industry – Sources of water for industries – Characteristics of industrial wastes – Industrial water requirements – quality and quantity

UNIT – II

Waste reduction – Volume reduction – Classification of wastes – Equalisation- Neutralisation – Floatation – Precipitation – Heavy metal removal - adsorption – Aerobic and anaerobic biological treatment – reed bed technology

UNIT - III

Measurement of industrial waste water flow – waste water characterization - Advanced waste reduction technologies – Ozonation – Membrane technologies – Ion exchange – Nutrient removal – recycling, reuse and resources recovery

UNIT – IV

Waste disposal methods- land treatment – water bodies, rivers, oceans – problems of disposal – Common effluent treatment plants- advantages and disadvantages – recirculation of industrial wastes – Effluent disposal methods – sludge treatment - disposal

UNIT – V

Manufacturing process and origin, characteristics, effects and treatment methods of liquid waste from different industries – steel – fertilizers – textiles – paper and pulp industries – oil refineries – coal and gas power plants- tanneries – sugar – textiles – distillery – dairy – food processing - distilleries

Text Books:

1. Eckenfelder, W.W., (2000)'Industrial Water Pollution Control'', Mc-Graw Hill.

Reference Books:

1. Industrial waste water treatment by A.D. Patwardhan.

GREEN FUEL TECHNOLOGIES
(Open elective)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course objectives:

Student shall be able to:

1. Understand various green fuel technologies available worldwide
2. Explain the production of Bio Ethanol from crops and agricultural waste
3. Understand the importance of biogas technology for methane production.

Course outcomes:

By the end of the course, the student should be able to:

1. Classify various types of green fuels [K2]
2. Examine the production of bio ethanol using fermentation process[K3]
3. Explain the generation of bio diesel using algae species[K2]
4. Interpret the production of bio diesel from plant seeds[K2]
5. Explain the production of biogas from biogas plant[K2]

SYLLABUS

UNIT-I**Introduction:**

Definition of Green Fuel, Classification, advantages and effects of green fuel-Plant based Biofuels- Thermochemical conversion of biomass to liquids & gaseous fuels.

UNIT-II**Bio Ethanol Production Technologies:**

Bio Ethanol from crops- cane sugar-Sugar Fermentation Process- Bio ethanol from starchy biomass- Bio Ethanol from agricultural waste- Banana pseudo stem as lignocellulosic substrate.

UNIT-III**Bio Diesel Production Technologies:**

Bio diesel from Algae: Algae culture-challenges-Algae culture for biodiesel production-biodiesel production with super critical fluid technologies.

UNIT-IV

Bio Diesel From Different Plant Seeds:

Chemical process -Palm oil diesel production- Bio diesel production from rubber seed oil & other vegetable oils.

UNIT-V

Biogas Production From Biomass:

Microbial production of methane(Biogas)- Biogas production by Biogas plant & other biogas technologies in India.

Text Books:

1. Biofuels Engineering Process Technology, Caye M. Drapcho, NghiemPhuNhuan, Terry H. Walker, McGraw-Hill, 2008.

Reference books:

- 1.Hand book of Plant Based Biofuels, Ashok Pandey, CRC Press, 2009.

GEOTECHNICAL ENGINEERING LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

The objectives of this course are to make the student aware of

1. Tests to identify the physical properties of soil.
2. Tests to identify the types of soil.
3. Tests to identify compaction characteristics and permeability of soil.
4. Tests to identify in-situ characteristics of soil.
5. Tests to determine shear strength of soil.
6. Tests to determine relative density of soil.
7. Tests to determine of CBR of soil.

Course Outcomes:

Students will be able to

1. Identify the physical properties of soil and classify various types of soil.(K2)
2. Determine the permeability of soil.(K3)
3. Determine compaction characteristics of soils and Estimate in-situ density of soil.(K3)
4. Determine the shear strength parameters of soils by various methods.(K3)
5. Estimate the California Bearing Ratio (CBR) of a soil.(K3)
6. Determine the relative density of a coarse-grained soil.(K3)

LIST OF PROGRAMS

1. Atterberg limits
2. Field density by Core Cutter method.
3. Field density by Sand replacement method
4. Grain size analysis (Sieve analysis)
5. Hydrometer analysis.
6. Specific gravity by pycnometer/density bottle method.
7. Permeability of soil – Constant headmethod.
8. Permeability of soil – Variable head method
9. IS light/heavy compaction.
10. CBR test/plate bearing test
11. Unconfined compression test
12. Triaxial compression test (u-u test)
13. Direct shear test
14. Vane shear test
15. Relative density.

***Atleast12experiments must be done**

Reference Book:

1. Soil Mechanics and Foundation Engineering by K.R. Arora

CONCRETE TECHNOLOGY LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int.Marks : 50****Ext. Marks : 50****Credits : 2**

Course Learning Objectives:

1. To test ingredients of concrete for assessing their physical properties.
2. To prepare and test concrete for studying fresh and hardened properties.

Course Outcomes:

Upon successful completion of this course, student will be able to

1. Conduct test and find consistency and fineness of cement.
2. Examine the specific gravity of cement.
3. Conduct test and determine the setting times of cement.
4. Determine the compressive strength of cement.
5. Determine the specific gravity of coarse aggregate and fine aggregate.
6. Determine the fineness modulus of coarse aggregate and fine aggregate.
7. Determine the bulking of sand.
8. Understand and determine workability of concrete by slump, compaction factor, flow table and Vee – Bee tests.
9. Evaluate hardened properties of concrete like compressive strength, split tensile strength and flexural strength.

LIST OF EXPERIMENTS**TESTS ON CEMENT**

1. Fineness of cement
2. Specific gravity of cement
3. Normal consistency of cement
4. Initial and final setting time
5. Compressive strength of cement for different grades of cement.

TESTS ON AGGREGATE

6. Specific gravity and unit weight of coarse and fine aggregates
7. Sieve analysis of coarse and fine aggregates and classification as per IS 383.
8. Bulking characteristics of sand

TESTS ON FRESH CONCRETE

9. Workability tests on fresh concrete by using:
 - (a) Slump cone
 - (b) Compaction factor apparatus
 - (c) Flow table
 - (d) Vee-Bee consistometer

TESTS ON HARDENED CONCRETE

10. Strength tests on hardened concrete

- (a) Compressive strength
- (b) Split tensile strength
- (c) Flexural strength

Reference Books:

1. Concrete technology manual by M.L.Gambhir

EMPLOYABILITY SKILLS
(Common to all Branches)

Tutorial : 3 Periods (VA-2+QA-1)
Exam : 3 Hrs.

Int.Marks : 30
Ext.Marks : 70
Credits : 1

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a 'Statement of purpose', 'Letters of Recommendation', business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron's GRE guide.
3. Newspapers like 'The Hindu', 'Times of India', 'Economic Times.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and PushpaLatha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II:Puzzle test, blood Relations, permutations, Combinations and probability Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III:Periods,Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

BASIC CODING
(Common to CE & ME)

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 1

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:**At the end of the course students will be able to**

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS**UNIT I Review of Programming constructs**

Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers

Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes

Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.

Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations Vs pointed locations, Referencing a 2D-Matrix.

UNIT IV User-defined data types, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

Reference Books:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by RobatLaphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>